Cover sheet & declaration form

Assignment details

Student number	**19327439**
Module number	**ENGR7019**
Assignment number and assignment type (for example, Assignment 1 Case study)	**Draft final report submission for supervisor formative feedback, due by 5th September**

Use of digital and Artificial Intelligence (AI) tools

Which digital/AI tools did you use? (for example, Word, Library databases, Co-pilot, Research Rabbit, none)	**Word; Excel; Library databases; Mendeley**
What did you use the digital/AI tools for? (one sentence about each tool/ not applicable if none)	**Word**: Report writing and note taking. **Excel**: Data analysis and creation of tables. **Library databases**: Reference searching. **Mendeley**: In-text references and bibliography.

Declaration

Select yes or no for the statements below.

Yes I confirm that the submission below is my own work.

Yes Where I have used other sources of information, I have attributed them in citations and a reference list.

Yes I have followed the assignment instructions regarding use of AI.

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- University Moodle courses on Academic Integrity and Use of Artificial Intelligence.
- Centre for Academic Development advice on Artificial Intelligence.
- Academic Conduct Procedure Definition of misuse of AI (3.1i).
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OXFORD BROOKES UNIVERSITY

ENGR7019 Engineering Dissertation Final Report

Optimization of LTC (Low Temperature Combustion) IC engine and simulation, using Hydrogen fuel and control strategy.

Diego Pilares Gallego – 19327439

Programme:	**MSc Motorsport Engineering**
Module:	**ENGR7019: MSc Dissertation**
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Contents

List of figures/Diagrams

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List of Symbols

$\boldsymbol{\mu_i^*}$: Mean of absolute

$\boldsymbol{\sigma_i}$: Standard deviation

$\boldsymbol{d_i(X)}$: Elementary effect of parameter $\boldsymbol{i}$ at point $\boldsymbol{X}$

$\Delta_{\boldsymbol{i}}$: Increment applied to parameter $\boldsymbol{i}$

$\boldsymbol{r}$: Number of trajectories

$\boldsymbol{X}$: Input set of control parameters

$\boldsymbol{X^{(j)}}$: Parameter vector at the $\boldsymbol{j}^{\text{th}}$ trajectory

$\boldsymbol{G_t}$: Accumulated reward at time $\mathbf{t}$

$\boldsymbol{r_t}$: Immediate reward at time $\mathbf{t}$

$\boldsymbol{\gamma}$: Reward weight

$\mathbf{R^2}$: Statistical measure of model prediction reliability

Highlights

- Developed GT-Suite model of hydrogen-fueled LTC/CAI engine incorporating advance control features, including High-Pressure EGR, Variable Geometry Turbocharging and Variable Valve Timing and Lift.
- A Design of Experiments, combined with a Morris sensitivity analysis, was used to identify the most influential control parameters impacting engine performance and NOx emissions over a large engine operating range.
- Created and validated Multi-Layer Perceptron machine learning metamodels, trained on extensive simulation datasets, enabling efficient prediction of engine performance metrics and optimization of engine control strategies.
- Demonstrated optimization capabilities with improved torque and power output at low and medium engine speeds, together with reductions in peak in-cylinder temperatures and a reduction in minimum achievable NOx emissions.

2025-09-05

Optimization of Low Temperature Combustion Internal Combustion engine and simulation, using Hydrogen fuel and control strategy.

Diego Pilares Gallego
Oxford Brookes University

Abstract

This project focuses on the optimization of a hydrogen-fueled Low-Temperature Combustion (LTC) internal combustion engine using Controlled Auto-Ignition (CAI) and advanced machine learning control strategies. The primary objective is to achieve high thermal efficiency and ultra-low NOx emissions through systematic simulation-based optimization and machine learning control strategy development and optimization.

A comprehensive methodology was employed, combining GT-Suite 1D engine modeling, Design of Experiments (DOE) parametric studies, Morris sensitivity analysis, and machine learning optimization techniques. The engine model replicates a 4-stroke, 11.7-liter, inline 6-cylinder, hydrogen direct injection engine integrated with High-Pressure Exhaust Gas Recirculation (HP-EGR), Variable Valve Timing/Lift (VVT/VVL) for both intake and exhaust valves, and Variable Geometry Turbocharging (VGT) systems.

Key results demonstrate successful optimization of critical engine parameters across the operating range, such as boost pressure, EGR valve throttle, EGR intercooler bypass diameter and VVT/VVL. The Morris sensitivity analysis identified the most influential parameters affecting thermal efficiency and NOx emissions. A machine learning framework incorporating Multilayer Perceptrons (MLP) was developed to create a metamodel capable of real-time engine optimization.

The optimized control strategy achieved significant improvements in engine performance while maintaining ultra-low NOx emissions through precise HP-EGR control and advanced combustion phasing. The developed metamodel demonstrates computational efficiency gains over traditional optimization approaches, making it suitable for real-time engine control applications.

This research contributes to the advancement of hydrogen-fueled LTC/CAI engines by providing a simulation framework and intelligent control system that addresses the challenges of combustion stability, emissions reduction, and thermal efficiency optimization across multiple operating conditions.

Introduction

Background

Hydrogen-fueled Low-Temperature Combustion (LTC)/Controlled Auto-Ignition (CAI) represents an important advancement in internal combustion engines aimed at simultaneously achieving ultra-low NO_x emissions and high thermal efficiency. Unlike conventional spark-ignition engines, LTC/CAI relies on auto-ignition chemistry to initiate combustion at lower peak temperatures, minimizing NO_x formation. Hydrogen's unique properties present both opportunities and challenges for LTC/CAI operation. On one hand, hydrogen allows lean-burn operation and extended ignition delay for improved mixing and reduced soot, on the other, its low energy density and tendency for backfire and pre-ignition necessitate advanced mixture formation and charge-dilution strategies.

HP-EGR has emerged as one of the preferred methods to control peak temperature and NO_x emissions in hydrogen engines. However, excessive EGR rates can reduce Brake Mean Effective Pressure (BMEP) and raise hydrocarbon and CO emissions, highlighting a delicate trade-off. Variable Valve Timing (VVT) and Variable Valve Lift (VVL) strategies offer another control axis by lowering effective compression ratio to extend LTC operability. Direct injection further enhances power density and mixture stratification over carbureted or port injection systems but introduces engine control complexity.

Accurate modeling of hydrogen LTC/CAI demands simulation of combustion kinetics, gas exchange, heat transfer, and control strategy. Industry-standard tools such as GT-Power provide one-dimensional (1D) modeling of cylinder pressure, mass flow, and species formation, yet their fidelity in transient LTC/CAI operation remains under-validated. Real-time engine control further complicates implementation: Model-based calibration is labor-intensive, and computational demands hinder closed-loop optimization in embedded Electronic Control Units (ECUs). Machine Learning (ML) methods offer accelerated mapping of multidimensional parameter spaces and adaptive control, but their integration with physics-based simulations for hydrogen LTC/CAI has not been fully explored.

Aim

Development and optimization of a hydrogen-fueled LTC/CAI engine model through 1D simulation and control strategy implementation, achieving ultra-low NO_x emissions and maximizing thermal efficiency across multiple operating conditions.

Objectives

1. Literature Review: Conduct a comprehensive literature review encompassing hydrogen LTC/CAI combustion chemistry, HP-EGR with intercooling strategies, VVT/VVL control methodologies, and machine learning applications in engine control systems.
2. Advanced Engine Model Development: Construct a detailed GT-Power 1D model of a 4-stroke, hydrogen direct-injection engine, incorporating validated combustion kinetics models, comprehensive heat transfer mechanisms, HP-EGR system with intercooler bypass control, and VVT/VVL dynamics.
3. Systematic Parameter Optimization: Execute Design-of-Experiments (DOE) parametric investigations across critical control variables including boost pressure, EGR valve position, intercooler bypass diameter, and valve timing/lift parameters, employing Morris sensitivity analysis to quantify parameter interactions.
4. Machine Learning Framework Development: Develop and train advanced machine learning regression models, using comprehensive DOE datasets to predict optimal control parameters for specified operating points.

Literature Review Summary

Hydrogen fueled engines operating with LTC/CAI and HP-EGR strategies have recently emerged as a viable option to achieve high thermal efficiency and low NOx emissions. LTC has been the focus of recent research, representing an advancement in internal combustion engines, offering potential improvements in both efficiency and emissions reduction. For air-fuel mixture combustion, low-temperature chemistry plays a crucial role in CO formation, with the induction period at low temperatures being significantly longer than in conventional engines (Soltesova et al., 2024). This extended induction period presents both challenges and opportunities for combustion control. Multiple options to facilitate CAI combustion have been investigated (Veza et al., 2023) including direct intake charge heating, higher compression ratios, more auto-ignitable fuels, and recycling of burnt gases (EGR). Among these approaches, EGR has been identified as the most appropriate method for 4-stroke SI engines transitioning to LTC/CAI operation.

HP-EGR systems extract exhaust gases before they reach the turbine and reintroduce them downstream of the compressor, with the addition of intercooling to reduce the temperature of the gases (Lu et al. 2024), enabling better control over combustion phasing and NOx emissions, particularly in lean conditions (Dyuisenakhmetov et al. 2025). The mechanism relies on H2O2 formation kinetics, where auto-ignition initiates at 1,000 K through exothermic reactions accelerated by EGR-diluted mixtures. It has been demonstrated that HP-EGR rates in the range of 20–35% are effective for hydrogen CAI combustion, maintaining peak in-cylinder temperatures below 1,650 K and reducing NOx emissions at lean conditions (Veza et al., 2023). This EGR configuration is critical for hydrogen engines, as hydrogen´s high flame speed and wide flammability range can otherwise lead to pre-ignition and increased NOx formation, especially at higher loads and stoichiometric mixtures (Kruczyński et al. 2019). Research in CAI combustion highlights the role of Variable Valve Timing (VVT) in managing residual gas fractions, with symmetric valve profiles enabling faster combustion than asymmetric timing due to enhanced thermal stratification (Cao et al. 2004).

Three main categories of hydrogen fuel delivery systems have been identified in the literature: carbureted injection, port injection, and direct injection [1]. Direct injection systems are further categorized into Low-Pressure Direct Injection (LPDI), which injects fuel when the intake valve closes during low pressure in the cylinder, and High-Pressure Direct Injection (HPDI), which injects fuel at the end of the compression stroke (Faizal et al., 2019).

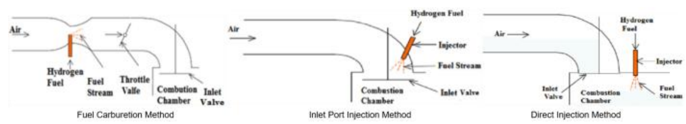

Figure 1. Hydrogen Fuel Induction Techniques (Faizal et al., 2019)

Direct injection approaches are more technologically advanced, involving mixture formation inside the combustion cylinder. The power output of a direct-injected hydrogen engine is reported to be 20% more than a petrol engine and 42% more than a carbureted engine (College of the Desert 2001).

While this approach shifts the pre-ignition problem from the manifold to the combustion chamber, it also reduces mixing time, potentially leading to non-homogeneous air-fuel mixtures with higher NOx emissions and requiring greater fuel rail pressure.

Control strategies for LTC/CAI hydrogen engines must address combustion stability across transient loads. VVT and Variable Valve Lift (VVL) strategies, especially retarding the Intake Valve Closing (IVC), can reduce the effective compression ratio and delay ignition, extending the engine's operation range (Zhang et al., 2007). The control of EGR flow and temperature is crucial, as excessive EGR rates can lead to incomplete combustion, higher hydrocarbon (HC) and carbon monoxide (CO) emissions, and reduced brake thermal efficiency (Park et al. 2015). Increasing EGR rates up to 20% can reduce NOx emissions by approximately 25%, but also increases particulate emissions by around 15%, highlighting a trade-off that must be managed through precise EGR valve control and cooling (Kikuchi et al., 2022).

Recent research (Fu et al., 2019) remarks that the combination of boost pressure and EGR can extend the load boundaries in CAI 4-stroke engines, highlighting the use of Variable Geometry Turbocharging (VGT) systems with intercooling for lower engine speeds.

Hydrogen´s high auto-ignition temperature (850 K) makes it challenging to achieve self-sustained combustion (Faizal et al., 2019), necessitating advanced control strategies such as Variable Valve Timing/Lift (VVT/VVL), direct injection, and EGR. CAI combustion is governed by the formation of hydrogen peroxide (H_2O_2) (College of the Desert 2001), with ignition typically occurring when the mixture reaches approximately 1,000 K. This temperature threshold provides a critical reference for predicting ignition timing and phasing in hydrogen CAI systems, and recent research has shown that EGR is the most effective strategy for controlling combustion phasing in 4-stroke engines transitioning to LTC operation. When SI engines are fueled with hydrogen only, NOx emissions increase, and the flame often retreats to the inlet system (Kruczyński et al. 2019). This backfire tendency, although possible to be reduced by injecting a poor mixture immediately after the inlet valve, represents one of the primary challenges in hydrogen engine development. This backfire tendency can be partly attributed to hydrogen's small quenching distance and minimal ignition energy requirement (0.02mJ compared to 0.24mJ for traditional gasoline-air mixtures) (Faizal et al., 2019).

The integration of simulation tools, particularly GT-Suite for this project's development, has become essential for the development and optimization of hydrogen-fueled LTC/CAI engines. These simulation capabilities are critical for validating engine models and optimizing control strategies before physical testing, thereby reducing development time and cost. Design of Experiments (DOE) modules facilitate parametric sweeps and sensitivity studies via multiple methods, such as the Morris method or the Sobol method (Tsvetkova and Ouarda 2021).

Machine learning (ML) has become increasingly important for real-time optimization and control of hydrogen LTC/CAI engines. Multilayer perceptrons (MLP) can predict optimal parameter values across operating conditions, reducing calibration errors (Tsvetkova and Ouarda 2021).

Although simulation approaches have advanced, a gap remains in the comprehensive validation of hydrogen-fueled LTC/CAI engines across multiple operating conditions. Most current optimization approaches are computationally intensive and unsuitable for real-time engine control. Development of simplified but accurate models capable of real-time optimization of LTC/CAI hydrogen-fueled engines is needed. The literature indicates potential for advanced injection strategies to optimize hydrogen utilization, but systematic exploration of these strategies specifically for LTC operation is limited. Current LTC engines with hydrogen face significant operating range limitations, with challenges at both low and high engine loads.

In summary, current research establishes that HP-EGR with intercooling, when combined with advanced LTC/CAI strategies and ML-based control, is a validated and effective approach for optimizing hydrogen-fueled engines. However, challenges remain in accurately modeling hydrogen-specific combustion phenomena, integrating ML controllers for real-time operation, and developing standardized validation metrics. This project aims to solve these gaps by utilizing industry standard simulation tools and ML methods, to deliver a robust and efficient control framework for hydrogen LTC/CAI engines.

Originality and Contribution

This research makes several unique contributions to hydrogen LTC/CAI engine technology development. The development of a GT-Power 1D engine model and its integration with hydrogen LTC/CAI kinetics, comprehensive HP-EGR, intercooler, and VVT/VVL systems provide unprecedented detail in published hydrogen combustion simulation.

The systematic application of a DOE-based Morris sensitivity analysis to multi-dimensional parameter optimization allowed the quantification of interactions between control variables and performance metrics, providing insights into parameter interdependencies.

The development of ML-guided control algorithms trained on highly accurate simulation data addresses the critical gap between advanced hydrogen engine concepts and practical real-time control algorithm implementation. The integration of MLP metamodels with physics-based simulations creates a computationally efficient framework, representing a significant advancement over existing computationally intensive approaches.

Methodology

Methodology summary

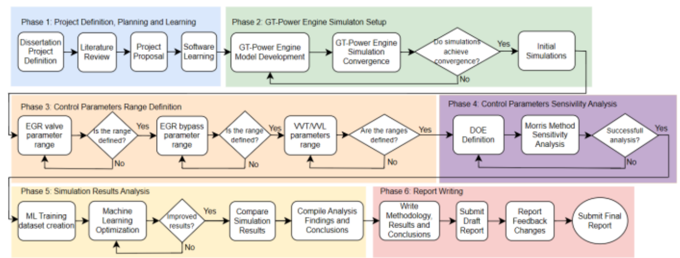

Figure 2. Project's methodology flowchart

The project methodology is structured into six distinct phases [2], where each phase includes specific tasks designed to each of the steps necessary to model, simulate and optimize a hydrogen-fueled Low Temperature Combustion (LTC)/Controlled Auto-Ignition (CAI) engine using GT-Power for engine modelling, simulation and Machine Learning (ML) control strategies development and optimization. This iterative approach involves conducting an initial operational range study of critical parameters, performing a DOE sensitivity analysis, and using ML to optimize High-Pressure Exhaust Gas Recirculation (HP-EGR), Variable Valve Timing (VVT) and Variable Valve Lift (VVL) control strategies.

Engine simulation setup

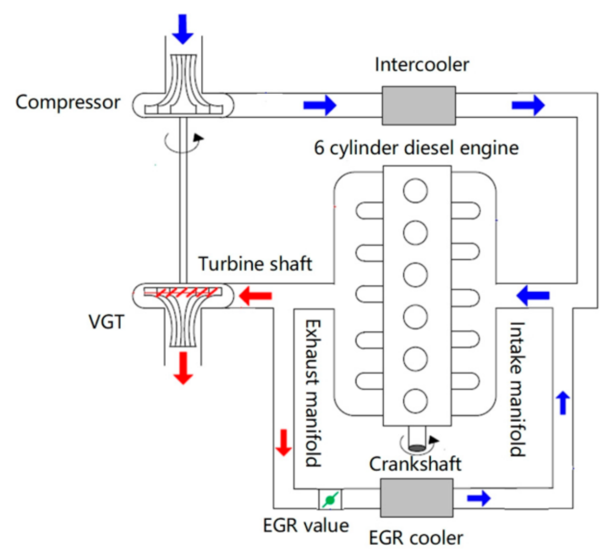

Figure 3. VGT-equipped engine with an EGR system. (Hu et al. 2019)

The GT-Suite model replicates a 4-stroke, 11.7 Liter, inline 6-cylinder hydrogen Direct Injection (DI) engine [3]. The engine model features individual cylinder models, with dimensions specified in Table 1.

Each cylinder incorporates three valves per cylinder (one intake, two exhaust) and dedicated intake and exhaust runners connected to common manifolds, enabling individual cylinder control while maintaining realistic gas dynamics. The combustion model enables a hydrogen single zone knock prediction model and the extended Zeldovich mechanism for NOx prediction.

Table 1. Engine model cylinder geometry input parameters

Parameter	Value	Units
Bore	119	mm
Stroke	175	mm
Connecting Rod Length	300	mm
Compression Ratio	12:1	
TDC Clearance Height	2	mm

The HP-EGR system extraction point is located upstream of the turbine housing. Exhaust gases are captured here and recirculated through an intercooler, which reduces temperatures based on coolant flow rates and heat exchanger effective parameters, and an intercooler bypass. The EGR valve is modulated to allow for different flow rates, depending on the engine load and Brake Mean Effective Pressure (BMEP) targets. The gases are reintroduced downstream of the compressor to ensure they mix with fresh air before entering the intake manifold.

The turbocharging system comprises a Variable Geometry Turbine (VGT) [4] and a centrifugal compressor. The turbine geometry is adjusted based on engine RPM and target BMEP, to optimize EGR pressures across load conditions, critical for maintaining an effective EGR at low loads.

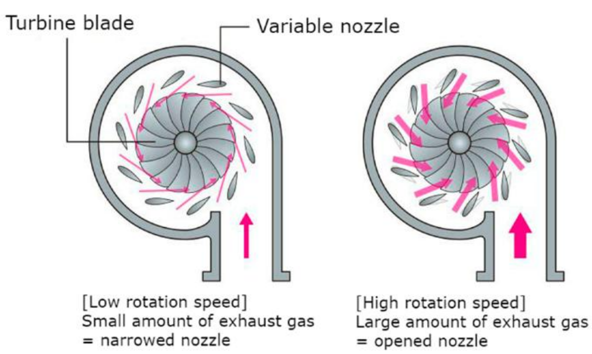

Figure 4. Variable geometry turbocharger (VGT) operating principle (Hu et al. 2019).

Control parameters range definition

This phase involves a focused case study to define operational ranges for the critical control elements identified for the project, EGR valve throttle position, EGR intercooler bypass diameter, and VVT/VVL strategies.

For the EGR valve throttle position, simulation results are used to compare in-cylinder NOx formation at Exhaust Valve Opening (EVO), against achieved BMEP and engine performance metrics. EGR intercooler bypass diameter tests evaluate cooling effectiveness and its impact on pre-ignition, resulting in engine knocking, and its effect on in-cylinder NOx formation at EVO. The performance of asymmetric and symmetric valve profiles is compared across EVO and Intake Valve Closing (IVC) timings, while the effect of different valve lifts on both the intake and exhaust valves is tested. These case studies establish preliminary parameter ranges without imposing fixed values, thereby ensuring flexibility for subsequent optimization stages.

Control parameters sensitivity analysis

The following phase utilizes a DOE sensitivity study, where the Morris screening method is employed to efficiently identify dominant parameters. The Morris method computes elementary effects across multi-level trajectories using the elementary effect for each parameter ($d_i(X)$) [Equation 1], the mean of absolute (μ_i^*) [Equation 2] and standard deviation (σ_i) [Equation 3] across r trajectories.

$$d_i(X) = \frac{f(X + \Delta_i) - f(X)}{\Delta_i}$$

Equation 1. Elementary effect of Morris

$$\mu_i^* = \frac{1}{r}\sum_{j-1}^{r} \left| d_i(X^{(j)}) \right|$$

Equation 2. Improved sensitivity measure based on the Morris method

$$\sigma_i = \sqrt{\frac{1}{r-1}\sum_{j=1}^{r} (d_i(X^{(j)}) - \mu_i)^2}$$

Equation 3. Standard deviation of elementary effects of Morriss

Based on these equations, μ_i^* quantifies the average influence of a parameter i on outputs, while σ_i indicates non-linear interactive effects between parameters.

Dataset generation

After the sensitivity analysis is performed, parametric sweeps are carried out across the defined operating ranges to systematically explore the engine's behavior in various scenarios. The simulation scenarios proposed for this project covers engine speeds between 1,000 and 5,500 rpm at wide open throttle, systematically varying the boost pressure, EGR valve opening angle, EGR intercooler bypass diameter, and intake and exhaust valve timing and lift.

For each simulation, engine performance and emission metrics are recorded, including NO_x at exhaust valve opening (EVO), Brake Mean Effective Pressure (BMEP), volumetric efficiency, air-fuel ratio (AFR), EGR rate, combustion temperature, and combustion stability. Only those simulation cases that achieve convergence within a 5% error threshold are retained for further analysis, as recommended in recent simulation studies (Rao et al. 2021).

This approach ensures the creation of a comprehensive, high-quality simulation dataset that is robust against nonlinear parameter interactions and representative of a wide range of engine operating conditions. The resulting database forms the basis for more detailed analysis, evaluation of parameter trade-offs, and efficient training of machine learning metamodels.

Machine learning control strategies optimization

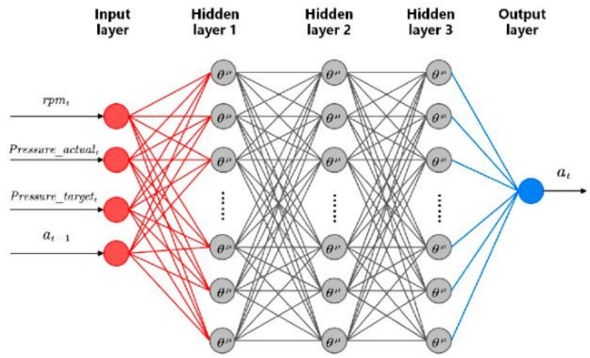

Figure 5. MLPs with hidden layers. (Hu et al. 2019)

Machine learning integration occurs through a multi-stage optimization framework. Initially, surrogate models based on multilayer perceptrons (MLPs) with hidden layers [5] replace conventional lookup tables for parameter value management. The previously developed dataset is then implemented with the experiments divided into training, validation and testing sets, with the split consisting of 70%, 15% and 15%, respectively, to ensure robustness and prevent overfitting. The use of MLP metamodels enables efficient evaluation of non-linear interactions between control parameters and engine performance and emission metrics, ultimately reducing computational costs while maintaining accuracy. This approach is particularly relevant to this project, as it enables rapid exploration and optimization of control parameters that would be computationally prohibitive using physical simulations, facilitating the development of robust control strategies. By leveraging ML metamodels, the project achieves scalable and advanced optimization that would not be possible with conventional methods (Norouzi et al. 2022).

Afterwards, a Reinforcement Learning (RL) control framework is used. Deep Deterministic Policy Gradient (DDPG) is used for dynamic engine operation optimization. The reward function must be configured to reward improvements in the engine's thermal efficiency and power output while penalizing deviations in NOx emissions, with different reward weights given for each parameter [Equation 4], where r indicates a reward, γ indicates a reward weight, or discount factor, and t indicates a certain moment (Hu et al. 2019).

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma r_{t+2}$$

Equation 4. Reinforcement Learning reward function

Once the ML model has been trained and validated, the simulations results are compared against the original simulations regarding NOx formation at EVO, engine thermal efficiency and overall engine performance.

Project limitations, assumptions and alternatives

To satisfy the project's objectives in a timely manner, some limitations [Table 2] and assumptions [Table 3] were adopted, while maintaining a balance between accuracy and practical constraints. Additionally, some alternative approaches [Table 4] were evaluated.

Table 2. List of limitations

Approach	Limitations	Mitigation
Simulation modelling	Simplifications/Assumptions in models	Refine models with additional validation data from different sources
Complex modelling and control strategies	High computational costs	Use of simplified models or increase computational power to reduce runtime
Machine learning algorithms	Dependency on large datasets	Simplify ML model or increase datasets through simulation
Hydrogen combustion modeling	Limited LTC/CAI validation data	Use of multiple literature datasets
Engine experimental validation data	No experimental data is available for validation	Comparison of achieved results range against published validated data

Table 3. List of assumptions

Assumption	Impact of the assumption on outcomes
Engine model geometry and combustion parameters are valid without experimental validation.	Uncertainty in baseline NOx emissions and engine thermal efficiency and performance.
No pressure losses from pipe bends.	Underestimating the pressure losses can increase the perceived EGR flow rate and boost pressure.
Engine cylinder temperatures are the same across the entire surface of each object (walls, head and piston).	Temperature gradients in the cylinder are not captured, potentially affecting combustion stability.
Homogeneous mixtures across the model.	Ideal mixtures can lead to lower emissions and higher combustion stability.

Table 4. List of alternatives

Task	Original approach	Alternative approach	Justification for the choice
Engine modelling	GT Power	AVL Boost	GT Power is chosen for better hydrogen combustion models and wider acceptance in hydrogen-fueled engine research.
Engine modelling	GT Power	Ricardo WAVE	GT Power is chosen for easier learning curve and wider acceptance in hydrogen-fueled engine research.
Sensitivity analysis	Morris method	Sobol method	Morris method is chosen as it is less computationally expensive.

Results

GT-Power engine model simulation setup

The 1D model required for simulation and optimization of the hydrogen-fueled LTC/CAI engine has been successfully developed within GT-Suite. The GT-Suite model [6] replicates a 4-stroke, 11.7 Liter, inline 6-cylinder hydrogen Direct Injection (DI) engine, with the addition of HP-EGR and VGT.

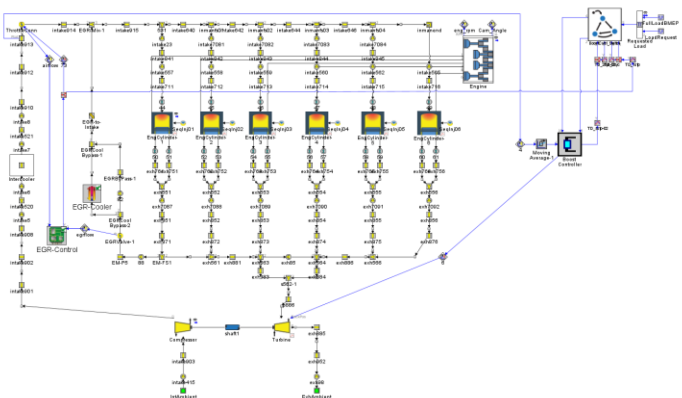

Figure 6. GT-Suite hydrogen LTC/CAI hydrogen engine model

The combustion model integrates the hydrogen knock prediction model and extended Zeldovich NOx emission model, critical for the later emissions, combustion stability and engine performance analysis. To regulate combustion temperature and emissions, an HP-EGR system was included. The system extracts exhaust gases upstream of the turbine, cooled via an intercooler, before reintroduction downstream of the compressor, where the exhaust gases mix with fresh air before entering the intake manifold.

The turbocharging system features a VGT, combined with a centrifugal compressor, designed for the thermodynamic properties of hydrogen combustion. The turbine geometry adjusts to engine speed and load to optimize EGR pressure and maintain effective EGR rates, especially at low speeds.

However, due to this engine model being developed without experimental validation data, the results achieved in the simulations differ significantly from what can be expected on a real engine. This factor is considered when discussing the following results and expected results for validated engines are discussed together with the achieved values.

Control parameters study: Effects of control variables

Systematic simulation sweeps were performed across the ranges defined for each identified control parameter at key rpm points, from 1000 to 5500 rpm every 500 rpm. For each sweep, engine performance data, such as Indicated Mean Effective Pressure (IMEP), brake torque, brake power, volumetric efficiency, indicated thermal efficiency, NOx emissions at EVO and maximum cylinder temperatures were recorded for trend analysis, allowing the definition of the control parameters range.

Boost pressure analysis

Boost pressure has proved critical for both power output and NOx emissions management. Increasing boost typically raise IMEP and brake torque, supporting higher power and efficiency [8], but also increases the maximum temperature achieved, thereby increasing NOx formation at higher pressures [7]. Lower boost values maintain lean air-fuel ratios, improving combustion stability and reducing NOx, at the cost of reduced power output.

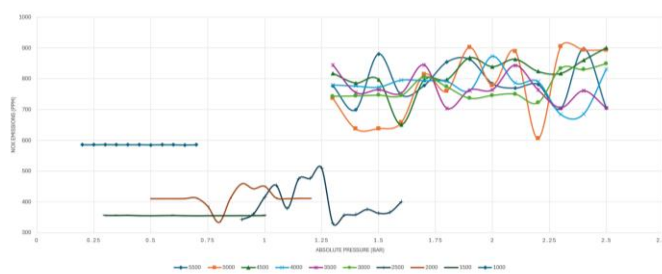

Figure 7. Boost pressure effects on NOx emissions

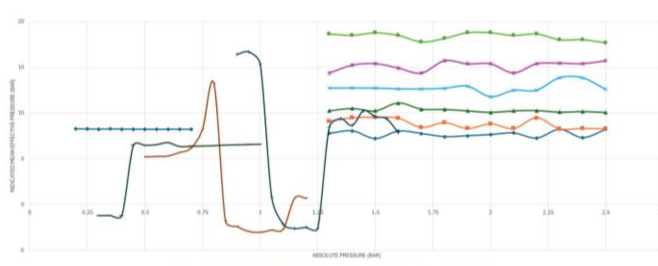

Figure 8. Boost pressure effects on IMEP

EGR Valve angle analysis

The EGR valve angle range is determined to be between 20 and 60 degrees at all rpms. Angles below 20° led to unstable combustion and misfire, while above 60° the reduction in volumetric efficiency outweighed any reduction on NOx emissions [9, 10]. Moderate angles in the 30º to 50° range achieved the best compromise, stabilizing combustion, maintaining high efficiency and reducing maximum temperatures, reducing NOx formation.

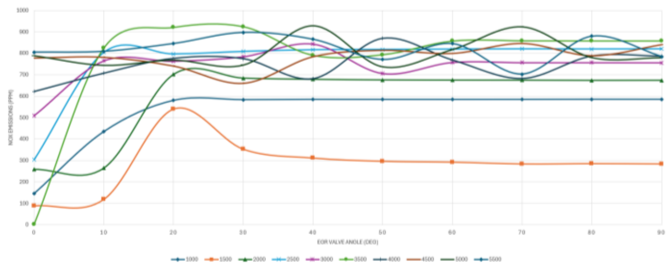

Figure 9. EGR Valve effects on NOx emissions

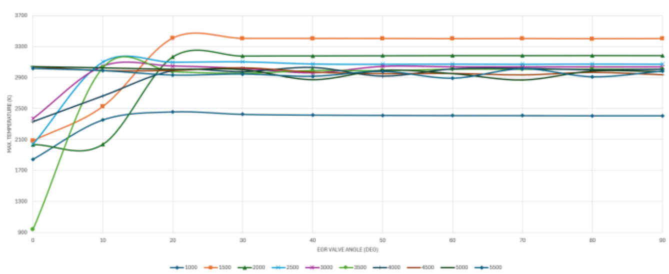

Figure 10. EGR Valve effects on in-cylinder temperature

These findings align with recent studies (Veza et al., 2023; Kikuchi et al., 2022), where it was found that that EGR rates above 25-30% are needed to improve combustion stability of hydrogen CAI engines in lean conditions.

EGR intercooler bypass analysis

The EGR bypass diameter was studied from 0mm to 50mm, or 0% to 100% of the diameter of the EGR piping [11, 12]. Its determined that the EGR Bypass diameter must not be smaller than 20mm at any point, as this causes consistent misfire across the rpm range. In events where misfire doesn't occur, NOx emissions are accentuated.

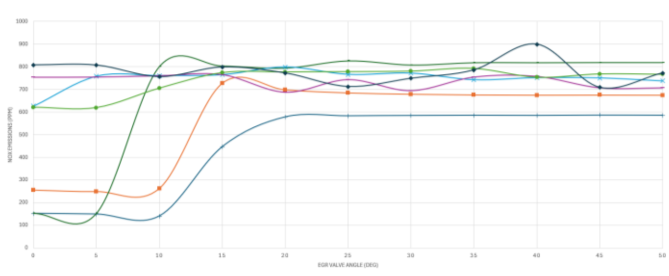

Figure 11. EGR intercooler bypass effects on NOx emissions

Meanwhile, the diameter should not be bigger than 45mm at any point, as this increases the in-cylinder temperature and reduces the volumetric efficiency, while worsening NOx emissions in the high rpm range. In the lower rpm range no benefit is seen in going over 30mm and it is encouraged to use lower values due to higher engine performance metrics and lower NOx emissions.

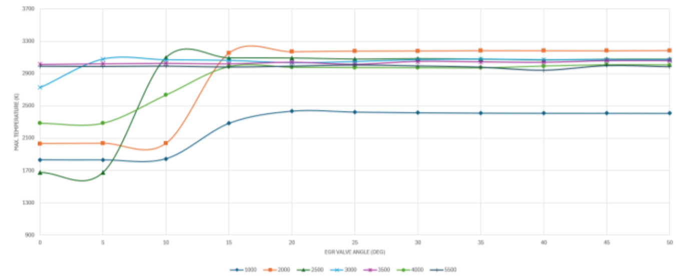

Figure 12. EGR intercooler bypass effects on in-cylinder temperature

VVT/VVL analysis

Variable Valve Timing/Lift sweeps revealed that both symmetric and asymmetric timing/lift combinations strongly affected combustion phasing, thermal efficiency, and NOx emissions. Results demonstrated that lift values used by a hydrogen-fueled engine are lower than those expected for conventional fuel engines and, to maintain air-flow rate, enabling higher loads and engine speed, valve lift needs to be increased for both intake and exhaust valves through the rpm range [13, 14], aligning with recent research (Faizal et al., 2019).

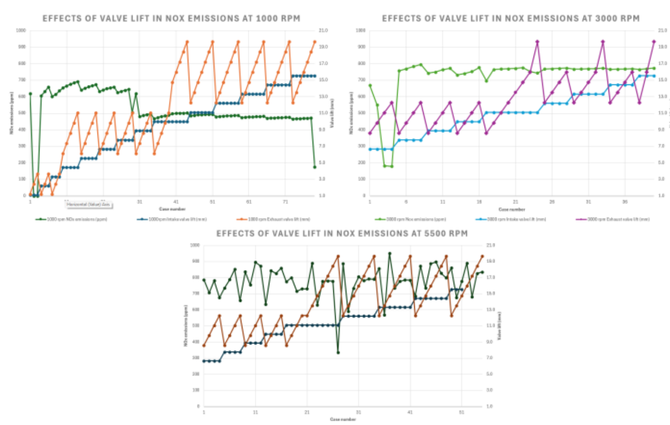

Figure 13. Effects of valve lift on NOx emissions

At high rpms, low lift values (intake and exhaust lower than 12.21mm and 15.97mm, respectively) risked misfire and efficiency loss, while excessive lifts (intake and exhaust higher than 15.55mm and 19.66mm, respectively) elevated peak temperatures and amplified NOx emissions.

At low rpms, low lift values (intake and exhaust lower than 3.33mm and 3.69mm, respectively) caused misfires due to insufficient air intake, while excessive lifts (intake and exhaust higher than 7.77mm and mm, respectively) consistently misfired due to reduced in-cylinder temperatures and excessive airflow [13, 14].

The results revealed that valve timing must be kept symmetrical with injection timing throughout the operating range to reduce peak in-cylinder temperatures and promote homogeneous mixtures, thereby reducing NOx emissions, as demonstrated in recent research (Li et al. 2021).

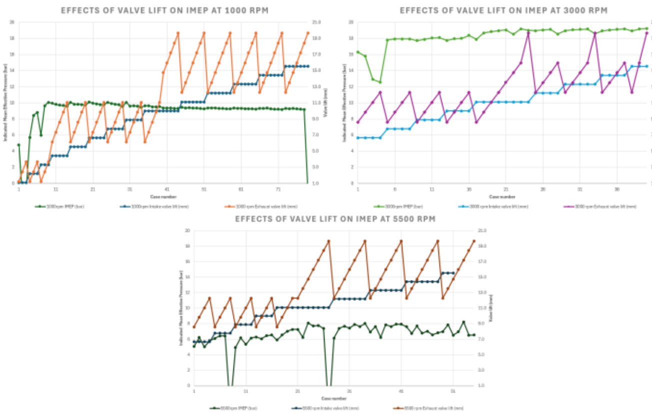

Figure 14. Effects of valve lift on IMEP

The defined operating range for the control variables in this study can be found in the appendix.

Multi-parameter DOE Morris sensitivity analysis

The Morris sensitivity analysis evaluated the impact of the control variables on engine performance and NOx emissions across the 1,000 to 5,500 rpm range.

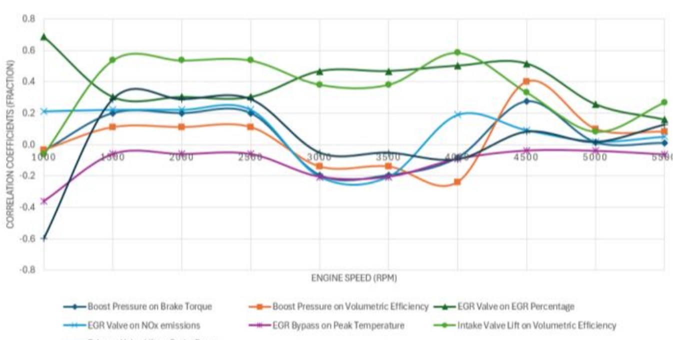

Figure 15. Correlation coefficients between main factor/response interactions

Correlation coefficients results [15] show that boost pressure consistently has the strongest influence on brake torque, power, and volumetric efficiency, especially at higher engine speeds, given its effect on air intake density and combustion stability but also showed adverse effects on peak temperatures and NOx emissions.

EGR valve position notably affected NOx emissions and temperature, due to its charge-dilution effect, confirming its role on thermal management. The EGR bypass diameter displays a moderate impact on volumetric efficiency and peak temperatures, reflecting its effect cooling the recirculated exhaust gases.

Intake and exhaust valve lifts display greater impact at lower engine speeds, affecting volumetric efficiency and combustion stability, helping regulate airflow and manifold pressures and thus facilitating EGR.

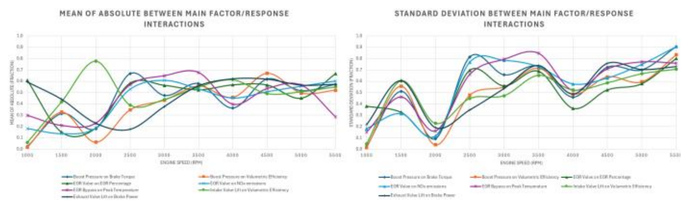

Figure 16. Mean of absolute and standard deviations of main interactions

The analysis of μ_i^* [16] identified the most dominant parameters for the optimization process, highlighting EGR valve and EGR intercooler bypass diameter as the most influential for NOx emissions reduction and boost pressure and intake and exhaust valve lifts as the most influential for engine performance. The analysis of σ_i [16] confirmed the non-linear interactive effects between the control variables, especially as the engine speed increases, supporting the need for machine learning metamodels, to capture these dynamics and allow for precise parameter optimization.

ML metamodels performance

The dataset used for the metamodels development in this study was generated through systematic parametric sweeps of the engine model in GT-Suite. Simulations covered a range from 1,000 to 5,500 rpm in 250rpm steps at wide open throttle, varying key control parameters such as boost pressure, EGR valve angle, EGR intercooler bypass diameter and intake and exhaust valve lift.

Simulation cases that converged within a 5% error threshold were retained, resulting in a comprehensive 12,240 cases dataset that captures nonlinear interactions between control variables and engine performance metrics. This dataset is partitioned into training, validation and testing sets, with the split consisting of 70%, 15% and 15%, respectively, achieving a robust development of the machine learning metamodels and preventing overfitting.

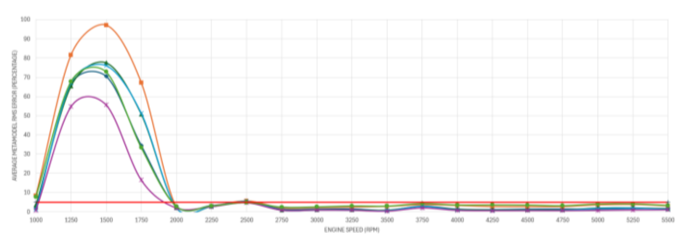

Figure 17. Average metamodel RMS Error

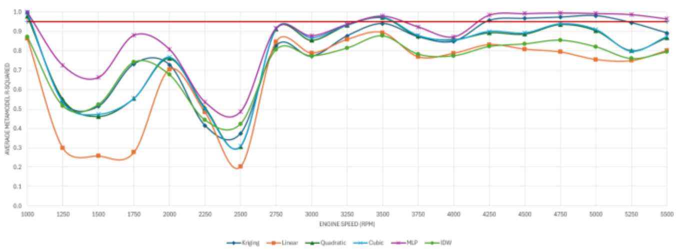

Figure 18. Average metamodel R-Squared

Initially, metamodels available in GT-Suite, such as Kriging, linear, quadratic, cubic, MLP and IDW (Inverse Distance Weighting) metamodels, were trained and compared. Performance for each target response was evaluated and then averaged across outputs for each case for concise analysis representation.

At lower engine speeds, all metamodel types show high RMS (Root Mean Squared) error [17], preventing their real use for precise engine control optimization, on the medium to high engine speeds, the 5% error threshold imposed is surpassed by all the trained models.

The R^2 values achieved at low and medium engine speeds [18] revealed a lack of confidence in all models except the MLP metamodel, as the 0.95 R^2 threshold is only surpassed by said metamodel.

At lower speeds, RMS Error values are above threshold for the brake torque, brake power, NOx emissions and peak temperature, R^2 values fall below threshold for all target responses. At medium to high speeds, RMS Error values are above threshold for the peak temperature, R^2 values are below threshold for the AFR and peak temperature.

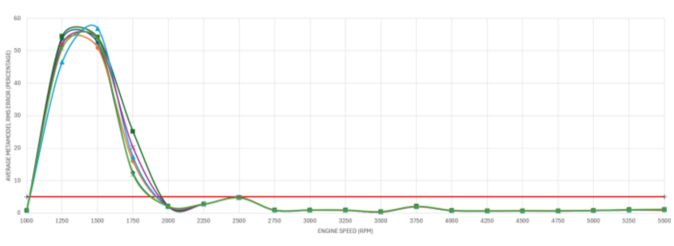

Figure 19. Average MLP metamodel RMS Error

Given its superior performance when compared to other models, the MLP metamodel received further training. Metamodels with 2 and 3 hidden layer configurations, using 10 neurons for the first hidden layer and 5 neurons for subsequent layers, were investigated to find the balance between metamodel accuracy and computational efficiency. Multiple training iterations are used in the training of the metamodels to address the natural variability in predictive models. The imposed RMS error threshold is surpassed by the MLP models in the 1,250 to 1,750 rpm range [19], while the R^2 threshold is surpassed in lower and medium engine speeds [20].

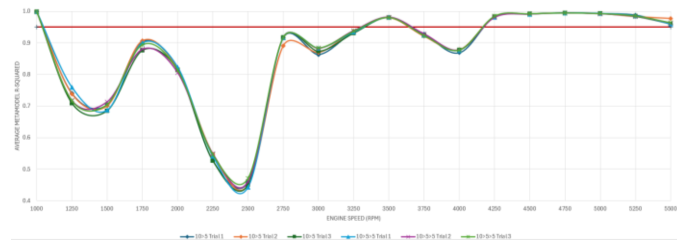

Figure 20. Average MLP metamodel R-Squared

For the individual target responses metamodel analysis, differences in R^2 were not substantial enough to solely determine metamodel selection, RMS Error provided a decisive metric for choosing the final metamodels used in the optimization process, summarized in Table 5.

The metamodel selection strategy used ensures accuracy and consistency across the target responses metamodels, supporting the reliability of the optimization process outcomes.

Table 5. Selected MLP metamodel performance per response

Response	MLP Metamodel	R^2	RMS Error
Brake Torque	10>5>5	0.883	24.404%
Brake Power	10>5	0.873	4.455%
Vol. Efficiency	10>5>5	0.957	0.012%
EGR Rate	10>5	0.940	0.538%
Air-Fuel Ratio	10>5	0.732	0.068%
NOx at EVO	10>5	0.911	4.95%
Peak Temperature	10>5	0.817	14.34%

ML optimization outcomes

The ML optimization was performed using a reward based optimization strategy, where different response weights and objectives were applied to each control parameter [Table 6].

As the main objective of the project is to reduce NOx emissions, results of the optimization process were analyzed and then compared to the original results that displayed the lowest NOx emissions across the studied engine speed range. A constraint was put in place for the engine AFR, ensuring that the mixture remained lean across the study, prohibiting AFRs lower than the stoichiometric AFR for hydrogen, 34.33:1 based on mass (College of the Desert 2001).

Table 6. Optimization objectives and weights

Response	Objective	Response Weight
NOx at EVO	Minimize	2
Volumetric Efficiency	Maximize	1.4
Brake Torque	Maximize	1.3
Brake Power	Maximize	1.3
EGR Rate	Maximize	1.2
Air-Fuel Ratio	Maximize	1.1
Peak Temperature	Minimize	1

Table 7. Comparison between original and optimized factors

		1000	1250	1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500
Original Boost Pressure	bar	0.30	0.57	0.62	0.85	0.60	0.80	1.13	1.55	2.00	2.10	2.20	2.00	2.40	2.15	1.60	1.55	1.50	1.35	1.40
Optimized Boost Pressure	bar	0.31	0.70	0.68	0.57	0.64	0.80	1.33	1.70	1.99	1.79	1.83	1.98	2.27	2.13	1.80	1.47	1.60	1.37	1.41
		2.08%	23.45%	10.26%	-33.40%	6.72%	0.08%	17.53%	9.53%	-0.46%	-14.95%	-16.96%	-0.96%	-5.48%	-0.86%	12.69%	-5.44%	6.51%	1.63%	0.88%
Original EGR Valve	deg	20.00	40.00	25.00	60.00	33.33	50.00	20.00	46.67	60.00	60.00	60.00	46.67	20.00	60.00	20.00	20.00	20.00	33.33	20.00
Optimized EGR Valve	deg	20.05	50.58	27.51	57.75	40.02	56.08	59.80	59.96	59.60	59.76	58.50	55.57	59.62	39.16	51.59	56.42	59.82	50.31	59.64
		0.23%	26.45%	10.03%	-3.75%	20.05%	12.16%	198.99%	28.49%	-0.67%	-0.41%	-2.50%	19.08%	198.10%	-34.74%	157.97%	182.12%	199.11%	50.93%	198.20%
Original EGR-Bypass	mm	25.00	25.00	25.00	30.00	30.00	30.00	45.00	20.00	45.00	20.00	20.00	45.00	20.00	45.00	20.00	20.00	20.00	20.00	20.00
Optimized EGR-Bypass	mm	25.04	25.68	27.52	27.68	24.63	33.53	22.46	35.77	44.64	44.91	44.95	44.99	44.99	29.52	44.99	39.68	44.91	33.56	24.12
		0.17%	2.73%	10.10%	-7.74%	-17.91%	11.78%	-50.09%	78.87%	-0.81%	124.55%	124.74%	-0.03%	124.94%	-34.41%	124.95%	98.40%	124.53%	67.81%	20.60%
Original Intake Valve Lift	mm	7.77	8.88	7.77	9.44	11.10	13.33	15.55	14.44	11.10	15.55	15.55	12.22	11.66	11.80	13.74	12.08	12.22	12.22	13.33
Optimized Intake Valve Lift	mm	5.81	7.20	6.53	7.92	10.65	11.71	14.99	14.43	15.44	15.54	15.55	15.46	15.31	14.23	15.33	15.49	15.53	15.49	13.07
		-25.22%	-19.00%	-15.98%	-16.13%	-4.06%	-12.12%	-3.59%	-0.06%	39.08%	-0.06%	0.00%	26.58%	31.27%	20.61%	11.55%	28.24%	27.17%	26.82%	-1.89%
Original Exhaust Valve Lift	mm	3.69	8.60	8.60	7.37	8.60	9.22	12.29	11.06	12.29	19.66	19.66	19.66	19.66	16.18	19.66	19.66	19.66	17.00	19.66
Optimized Exhaust Valve Lift	mm	3.75	9.05	6.71	11.52	14.74	11.81	19.58	18.19	19.66	19.48	18.77	19.19	19.33	19.63	19.29	19.65	19.66	19.62	19.66
		1.82%	5.17%	-22.02%	56.29%	71.41%	28.18%	59.33%	64.47%	59.95%	-0.95%	-4.56%	-2.41%	-1.67%	21.31%	-1.90%	-0.05%	-0.03%	15.40%	0.00%

A comparison between original and optimized factors [Table 7] demonstrates the influence of the optimization process in the pursuit of the objectives set. Firstly, the evolution of the factors across the engine speed is noteworthy, following a more linear progression than the original values, consistent with the evolution of the factors that we might observe in a real engine. The optimized boost pressure and valve lifts show controlled, progressive increases, resulting in improved engine performance, efficiency and combustion stability. EGR parameters show an increase in their values at higher engine speeds, aiming for higher EGR rate values to reduce NOx emissions.

Although the NOx emissions [21] did not perceive an improvement post-optimization, resulting in slightly higher or similar emissions across the rpm range, a reduction in minimum achievable NOx emissions was achieved at 1250rpm, reducing NOx emissions from 325ppm to 277ppm.

Peak temperatures [21] were consistently reduced across the entire engine speed range, with an average reduction of 37K in peak temperature and a maximum reduction of 70K at 4,000 rpm. This reduction in peak temperature contributes to pre-ignition risk mitigation, combustion stability and engine reliability improvements.

Also noteworthy is the temperature peak produced at 1,500 rpm, resulting from the unreliability of the trained metamodel for that engine speed, where an R^2 value of 0.816 and an RMS error of 88.87% were observed.

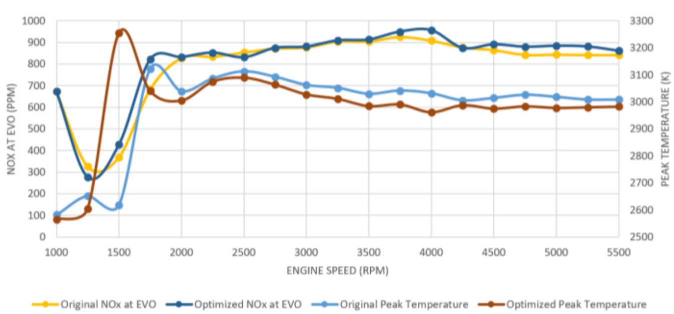

Figure 21. MLP optimized NOx emissions and peak temperature

Engine performance responses [22] achieved measurable improvements in all key indices, as volumetric efficiency was increased or maintained across the rpm range, peak efficiency remained at the 2,750rpm mark, where the compressor achieved an efficiency of 67%.

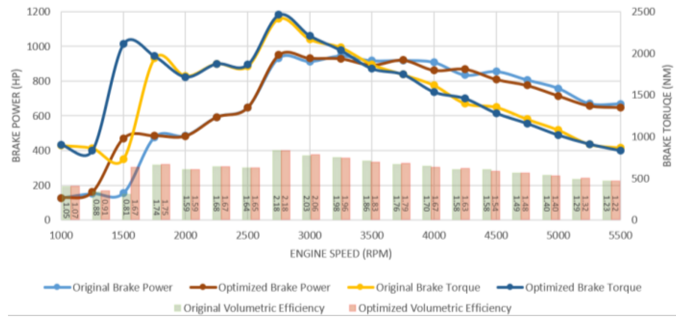

Figure 22. MLP optimized brake torque, power and volumetric efficiency

Brake torque and power displayed significant improvements at low and medium engine speeds, with average increases of 164Nm and 41HP, respectively, compared to the original results. In contrast, output at higher engine speeds displayed slight reductions, averaging 25Nm and 16HP less overall. In addition, peak values were improved, increasing torque from 2,415 to 2,460Nm and power from 947 to 950HP, both at 2,750rpm.

The EGR rate was consistently increased by the optimized results, with an average increase of 2.18%, reinforcing the optimization objective emphasis on NOx emissions reduction.

The AFR remained unchanged from the reference value, maintaining lean conditions at an AFR of 41:1, with no substantial deviation across the evaluated range.

Discussion

The results produced in this project provide valuable insights into the optimization of a hydrogen-fueled LTC/CAI with advanced machine learning control strategies optimization.

Despite the comprehensive simulation framework and systematic methodology implemented, the optimization outcomes revealed limitations in performance improvements, with performance metrics, such as NOx emissions reduction, brake torque and power, not meeting expectations and the achieved volumetric efficiency resulting unrealistic.

This discussion aims to critically analyze the key results, identify potential causes for errors and deviations, and contextualize findings against established literature and underlying assumptions.

Engine model simulation setup

The GT-Suite 1D model developed [6] successfully represents an 11.7 L inline-6 hydrogen DI LTC/CAI engine incorporating HP-EGR, VGT, and VVT/VVL control systems. However, as noted, the absence of experimental validation data introduces uncertainties, particularly in combustion parameters and thermal boundary conditions, that propagate through simulation outputs. Assumptions such as uniform cylinder wall temperatures and homogeneous mixture formation oversimplify real engine complexities, likely contributing to discrepancies in predicted combustion and emissions behavior.

A significant inconsistency observed is the abnormality in the pressure difference between the intake and exhaust manifolds, where the modeled intake pressures exceed the exhaust pressures at most operating points, something that is physically unfeasible in practical engine operation. This inversion is likely to limit the accurate calculation of EGR rates and flow dynamics, leading to an underestimation of effective EGR rates and, consequently, higher NOx emissions compared to validated engines. Sensitivity to EGR valve positioning, a critical factor in NOx emissions and in-cylinder temperature regulation, showed unexpected trends and a diminished influence at higher engine speeds, indicating non-linearities and possible limitations of the model in representing flow or combustion kinetics. These issues underscore the fundamental importance of experimental data for calibrating and verifying 1D models of LTC/CAI hydrogen engines.

Boost pressure effects

One of the most glaring issues identified throughout the project development is the prevalence of absolute pressures consistently below atmospheric pressure, with values dipping unjustifiably beneath this threshold across most of the engine operating range. Absolute pressures above 1 bar consistently caused misfires and simulation convergence issues in the 1,000 to 2,500rpm range, challenging fundamental thermodynamic principles governing turbocharged engine operation. Intake manifold pressures should not fall below atmospheric pressure at wide open throttle conditions in both turbocharged or naturally aspirated engines, implying the existence of erroneous engine modelling or compressor mapping or both.

The simulations deliberately excluded the compressor operating points from the highest efficiency region across the entire engine speed range, as this would drive the EGR rate to zero, resulting in an intake temperature insufficient to produce combustion of the mixture and, consequently, causing misfiring. This is due to the pressure at the intake manifold being higher than the pressure at the exhaust manifold, preventing exhaust gases from being transferred into the intake.

The Morris sensitivity analysis also demonstrated the simulations failure to capture appropriate flow interactions, as the correlation coefficient of boost pressure with volumetric efficiency, brake torque and power proved contrary to what may be expected, as the magnitude of the coefficient was negligible at the majority of situations and between 3,000 and 4,000rpm an increase in boost pressure showed negative impact on volumetric efficiency, brake torque and power.

EGR valve and intercooler bypass effects

EGR has been widely recognized as a critical parameter on combustion stability improvements and emissions reduction, for both soot and NOx, with the latter being the primary focus of this project. However, the simulation results noticeably misrepresented the true impact of EGR implementation on NOx emissions reduction, particularly evident in the Morris sensitivity study, where the correlation coefficients indicate an unexpectedly low effect of both EGR valve positioning and EGR intercooler bypass diameter on NOx emissions compared to what would be expected from a validated engine.

The fundamental role of HP-EGR implementation is to lower peak combustion temperatures, by diluting the combustion mixture with exhaust gases, and thereby reducing NOx emissions. For these reasons, a strong negative correlation between EGR valve positioning and NOx emissions is expected. Contrary to this, the simulation results show a small positive correlation, averaging to 0.08 across the engine speed range, peaking at a maximum of 0.22, results which appear to be counterintuitive, as a larger negative correlation (i.e. -0.7) is expected across the rpm range. However, as the engine speed increases, increasing the intake manifold pressure and naturally reducing the amount of exhaust gases that can be diluted into the mixture, the correlation coefficient magnitude is expected to decrease, fact which is well represented on the achieved sensitivity analysis results.

Additionally, the relation between EGR valve positioning and the EGR rate is also underrepresented in the performed simulations, as the correlation coefficient averaged to 0.396, indicating a moderate relation between them. The peak correlation coefficient of 0.685, achieved at 1,000rpm, would be the average correlation coefficient of the EGR valve positioning on the EGR rate one could expect on a validated engine model.

Regarding the EGR intercooler bypass diameter, its effects on peak in-cylinder temperatures exhibited inconsistencies. Larger diameters are expected to reduce exhaust gases passing through the intercooler, this reducing the cooling effect and achieving higher intake mixture temperatures and thus, higher peak in-cylinder temperatures. This effect should result in a positive large correlation coefficient between EGR bypass diameter and peak temperatures; however, the sensitivity analysis revealed a smaller negative correlation between them, averaging to -0.12 and peaking only at -0.36 at 1,000rpm. This suggests that contrary to what might be expected, increasing the bypass diameter is associated with a reduction in peak temperatures, which is inconsistent with the expected thermodynamic behavior explained above.

Valve lift control effects

Valve lift control strategies play a crucial role in regulating airflow into and out of the cylinder, having a direct impact on volumetric efficiency, combustion stability and NOx emissions. By adjusting the valve lift, the effective flow area can be controlled, influencing the AFR and in-cylinder temperature. Optimal valve lifts enable effective air-fuel ratio control and reduce peak in-cylinder temperatures, essential for minimizing NOx emissions. On lower engine speeds, valve lifts are expected to be smaller, to prevent misfires and elevated peak in-cylinder temperatures, as engine speed increases, larger valve lifts are expected to increase air flow to maintain lean AFR, while supporting higher loads and improving performance.

Simulation results showed an increase in valve lift as engine speed increased, as expected, with the same trend on the machine learning optimization results, however, the sensitivity analysis showed a higher implication of intake valve lift than exhaust valve lift on determining volumetric efficiency, brake torque and power, especially at higher engine speeds, where the correlation coefficient of the exhaust valve lift against brake power was completely negligible. The implications of valve lift on NOx emissions were not accurately represented in these simulations, as the correlation coefficient was mostly negligible and, when it was significant, it varied between favoring greater lift and advising against it in the following case studied.

Notably as well, valve timing, essential for controlling combustion phasing, in-cylinder temperature and NOx emissions, was fixed during the optimization phase, following findings in the initial parameter range study. This approach was justified by confirming symmetry in valve timing as optimal for LTC/CAI operation, consistent with recent literature. However, by not varying valve timing, the study reduced its dimensionality, making it more approachable, but limited exploration of potential improvements through combustion phasing adjustments, such as negative valve overlap or late intake valve closing.

ML metamodels performance and optimization limitations

Although the MLP metamodels demonstrated superior predictive performance compared to the alternatives, they still displayed significant RMS errors and reduced predictive confidence, especially at low to medium engine speeds, undermining control parameters optimization reliability. The worst example of metamodel performance is seen at 1500 rpm, with an average RMS error of 53%, reflecting limitations in capturing complex nonlinear interactions between the control factors and responses. These metamodel inaccuracies propagate into the optimization process, causing implausible control parameter adjustments and unreliable engine performance predictions.

However, these metamodels inaccuracies are not only due to modelling limitations within the MLP architecture but largely inherited from the underlying simulation dataset used for training, validating and testing the metamodels. The unrealistic parameter values used on the dataset and inaccurate results obtained in the dataset, especially on the 1,500rpm dataset, where large jumps between EGR rates and engine performance metrics arouse across the whole dataset, increasing the unreliability of the results.

These metamodel deficiencies translated into issues on the optimization process, where a weighted reward function was implemented. Different reward weights were tested, with the objective of NOx emissions reduction, with the best results achieved using the weights available in Equation 4, but given the flawed dataset, the reduction in NOx emissions could not be carried out, given the small effect the control parameters used in the training dataset had on charge dilution and thus, in NOx emissions. These problems can be once again observed on the results achieved by the optimization process for 1,500rpm, where in-cylinder temperature, brake torque and power are highly increased compared to the adjacent operating points and the original results achieved, discarding issues with how the reward function was implemented, suggesting that this function is theoretically and practically implemented correctly.

Conclusions

This project presented a comprehensive study on the optimization of a hydrogen-fueled Low Temperature Combustion (LTC) internal combustion engine using Controlled Auto-Ignition (CAI) and machine learning (ML) control strategies. Through detailed 1D modeling in GT-Suite, sensitivity analysis, and ML-based metamodel development, the project aimed to achieve ultra-low NOx emissions while maximizing thermal efficiency and engine performance across the engine operating range.

The developed GT-Suite 1D engine model represented a 4-stroke, 11.7-liter inline 6-cylinder hydrogen direct injection engine integrated with High-Pressure Exhaust Gas Recirculation (HP-EGR), Variable Geometry Turbocharging (VGT), and Variable Valve Timing/Lift (VVT/VVL) systems. Simulation case studies established relevant operational ranges for critical control parameters, such as boost pressure, EGR valve position, EGR intercooler bypass diameter, and intake and exhaust valve lift values. The use of a Design of Experiments (DOE) combined with a Morris sensitivity analysis enabled the identification of dominant parameters that influence engine performance and emissions, which, although not adequately representing the magnitude of the actual influence of these parameters, highlighted the fundamental role of boost pressure for engine performance and of the EGR valve and bypass diameter for NOx emissions control.

Machine learning metamodels based on Multilayer Perceptrons (MLPs) were trained and validated on a large dataset of over 12,000 simulation cases, capturing complex nonlinear interactions between control parameters and engine responses. These metamodels demonstrated superior predictive accuracy compared to other available machine learning metamodels, especially in the medium to high engine speeds. The optimization framework employed a weighted reward function prioritizing NOx emissions reduction alongside performance metrics, yielding improvements in volumetric efficiency, brake torque and power at low and medium engine speeds. Although the main project objective of NOx emissions reduction was not achieved across the studied engine speed range, the engine minimum achievable NOx emissions improved, achieving 277ppm at 1,250 rpm. Additionally, peak in-cylinder temperatures were consistently reduced, contributing to enhanced combustion stability, reducing pre-ignition risk and enhancing engine reliability.

Despite these results, the project faced inherent limitations, mainly due to the lack of experimental engine validation data and simplifications in the combustion models and thermal boundary condition models. Notable inconsistencies in intake and exhaust manifold pressures, EGR rates, and charge dilution effects, resulted in unrealistic reference values for both control parameters and responses, affecting accuracy in both the sensitivity and optimization analyses. The metamodels, while generally achieving good performance, exhibited reduced confidence and elevated errors at low engine speeds, primarily due to the underlying dataset shortcomings, rather than the metamodels architecture limitations. These data quality issues impacted the reliability of the optimization results, highlighting the crucial need for refined and validated simulation input data.

Symmetrical valve timing was utilized throughout the study, limiting the exploration of potential combustion phasing optimizations such as negative valve overlap, which has demonstrated promising results in other LTC engine research. Additionally, the ML optimization framework focused on steady-state engine operating conditions, without incorporating transient engine behavior under dynamic load conditions. The computational cost and convergence challenges constrained certain parametric ranges, restricting complete exploration of the full engine operating range.

To further develop and refine the findings presented in this dissertation, the following research opportunities are suggested:

- Experimental validation: Obtaining high quality experimental engine data would enable rigorous calibration and validation of the GT-Suite model and its combustion kinetics, improving the accuracy of the simulations in engine performance and emissions predictions. This will reduce the current uncertainties resulting from simplified assumptions and unvalidated parameters.
- Expanded control parameters: Introducing variations in valve timing and injection strategies as additional control parameters will facilitate a broader exploration of the effects of combustion timing, potentially reducing NOx emissions and improving volumetric efficiency.

- Transient and real-time control optimization: Extending the framework of machine learning metamodels to consider transient engine operating conditions, enhancing the applicability of optimized control strategies to real-world driving scenarios and dynamic load demands.
- Swarm optimization parallel to sensitivity analysis: Combining optimization algorithms such as particle swarm optimization with sensitivity studies could more comprehensively identify optimal combinations of control parameters beyond approximate optimal values achieved with the current sequential approach.
- Cylinder coating materials research: Investigating in-cylinder surface coating materials or gels to reduce heat transfer losses and moderate peak temperatures offers a promising way to improve thermal efficiency while decreasing NOx emissions.
- Larger and higher quality datasets: Generating expanded simulation datasets, including finer parameter sweeps and improved convergence criteria, will improve machine learning metamodels training and generalization, reducing errors, especially at low engine speeds.

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Definitions/Abbreviations

LTC	Low Temperature Combustion
CAI	Controlled Auto-Ignition
DOE	Design Of Experiments
HP-EGR	High-Pressure Exhaust Gas Recirculation
VVT	Variable Valve Timing
VVL	Variable Valve Lift
EVO	Exhaust Valve Opening
IVC	Intake Valve Closing
VGT	Variable Geometry Turbo
BMEP	Brake Mean Effective Pressure
IMEP	Indicated Mean Effective Pressure
AFR	Air-Fuel Ratio
LPDI	Low-Pressure Direct Injection
HPDI	High-Pressure Direct Injection
MLP	Multi-Layer Perceptrons
IDW	Inverse Distance Weighting
RMS	Root Mean Squared
R-Squared (R^2)	A statistical measure used to indicate confidence in the prediction of results from models. The better the linear regression, the closer the value will be to 1.

Appendix

Engine combustion object details

Chemistry file obtained from https://combustion.llnl.gov/archived-mechanisms/hydrogen.

Control parameters range details

Table 8. Control parameters defined range

	RPMs	5500		5250		5000		4750		4500		4250		4000		3750		3500		3250	
		Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper
Boost	bar	1.4	1.6	1.35	1.6	1.3	1.6	1.45	1.75	1.6	1.9	1.75	2.15	1.9	2.4	1.85	2.3	1.8	2.2	1.65	2.1
EGR Valve	deg	20	60	20	60	20	60	20	60	20	60	20	60	20	60	20	60	20	60	20	60
EGR Bypass	mm	20	45	20	45	20	45	20	45	20	45	20	45	20	45	20	45	20	45	20	45
Intake valve lift	mm	13.33	15.55	12.22	15.55	12.22	15.55	12.08	15.55	11.94	15.55	11.80	15.55	12.22	15.55	12.22	15.55	11.38	15.55	11.24	15.55
Exhaust valve lift	mm	15.98	19.66	15.67	19.66	15.36	19.66	15.05	19.66	14.75	19.66	14.44	19.66	14.13	19.66	13.52	19.66	12.90	19.66	12.60	19.66

	RPMs	3000		2750		2500		2250		2000		1750		1500		1250		1000	
		Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper
Boost	bar	1.5	2	1.25	1.7	1	1.4	0.8	1.2	0.6	1	0.525	0.85	0.45	0.7	0.3	0.7	0.3	0.7
EGR Valve	deg	20	60	20	60	20	60	30	60	20	60	20	60	25	60	40	60	20	60
EGR Bypass	mm	20	45	20	37.5	20	45	30	35	20	30	20	30	25	30	25	30	25	30
Intake valve lift	mm	11.10	15.55	11.10	14.44	11.10	15.55	9.99	13.33	7.77	11.10	6.11	10.55	5.55	7.77	4.44	8.88	3.33	7.77
Exhaust valve lift	mm	12.29	19.66	11.06	18.43	12.29	19.66	9.22	13.52	8.60	14.75	7.37	13.52	6.14	9.83	8.60	11.06	3.69	9.83

Boost pressure

A value highlighted in red indicates a misfire.

Table 9. Boost pressure analysis NOx emissions

rpm \ bar	2.5	2.4	2.3	2.2	2.1	2	1.9	1.8	1.7	1.6	1.55	1.5	1.45	1.4	1.35	1.3
5500	704.95	897.48	703.47	781.87	769.75	784.74	863.41	854.22	777.82	747.78		880.62		699.82		777.77
5250	667.47	889.39	892.48	887.15	693.53	894.78	891.30	772.50	771.02	883.61		884.17		892.78		672.26
5000	893.26	893.26	904.51	605.90	889.72	779.10	902.12	760.00	814.08	657.30		638.18		637.44		736.28
4750	841.32	665.28	622.35	860.10	780.64	865.32	643.09	750.78	763.32	780.78		740.87		722.58		624.36
4500	900.83	860.37	817.52	823.75	863.90	839.35	868.51	797.24	795.57	650.91		797.74		786.25		817.90
4250	813.81	755.13	755.00	764.02	757.49	783.28	788.09	784.71	777.14	778.65		771.80		776.17		777.17
4000	831.97	685.58	684.45	791.14	786.84	873.85	761.72	791.36	793.84	795.60		772.92		776.39		780.38
3750	861.03	775.43	702.94	774.23	768.17	702.05	861.51	857.99	775.25	775.33		703.20		865.16		769.03
3500	705.05	760.97	705.87	764.37	843.74	763.82	763.37	704.31	844.54	752.98		764.31		755.20		845.10
3000	848.83	829.57	833.59	722.36	750.08	746.03	736.84	773.92	806.02	745.20		746.71		744.13		743.28
2500										400.08	366.51	363.47	375.81	358.56	356.92	330.00
2000																
1500																
1000																

rpm \ bar	1.25	1.2	1.15	1.1	1.05	1	0.95	0.9	0.85	0.8	0.75	0.7	0.65	0.6	0.55	0.5	0.45	0.4	0.35	0.3	0.25	0.2
2500	510.13	476.23	475.02	379.30	454.34	415.58	360.17	342.97														
2000		410.70	410.82	409.91	412.02	450.04	442.04	458.23	411.98	333.30	385.25	412.34	410.10	409.84	409.80	409.82						
1500						355.51	355.37	355.37	355.29	355.33	355.30	355.15	355.28	355.52	355.42	355.24	355.34	355.72	355.58	355.60		
1000												585.33	584.55	585.43	585.69	585.13	585.87	585.41	585.64	586.34	585.51	585.55

Table 10. Boost pressure analysis IMEP

rpm \ bar	2.5	2.4	2.3	2.2	2.1	2	1.9	1.8	1.7	1.6	1.55	1.5	1.45	1.4	1.35	1.3
5500	8.19	7.30	8.21	7.25	7.81	7.63	7.51	7.40	7.76	8.01		7.19		8.04		7.76
5250	8.88	7.96	7.94	7.92	8.64	7.66	7.70	8.26	8.13	7.80		7.79		7.64		8.70
5000	8.27	8.32	8.28	9.45	8.31	8.81	8.31	8.93	8.42	9.42		9.51		9.46		9.11
4750	9.46	10.23	10.30	9.32	9.59	9.27	10.17	9.59	9.63	9.59		10.03		9.83		10.26
4500	10.06	10.16	10.10	10.27	10.23	10.07	10.24	10.39	10.42	11.08		10.23		10.48		10.24
4250	11.57	12.02	12.02	11.85	11.81	11.73	11.72	11.74	11.80	11.78		11.82		11.78		11.78
4000	12.56	13.83	13.84	12.48	12.45	11.76	12.91	12.68	12.61	12.61		12.70		12.72		12.70
3750	13.28	14.02	14.34	13.91	13.99	14.36	13.29	13.26	14.00	14.00		14.37		13.32		14.10
3500	15.71	15.40	15.45	15.38	14.39	15.38	15.37	15.71	14.37	14.91		15.40		15.23		14.38
3000	17.65	17.99	17.98	18.63	18.46	18.77	18.77	18.13	17.75	18.47		18.76		18.48		18.63
2500										7.86	9.37	9.62	10.27	8.66	9.40	8.45
2000																
1500																
1000																

rpm \ bar	1.25	1.2	1.15	1.1	1.05	1	0.95	0.9	0.85	0.8	0.75	0.7	0.65	0.6	0.55	0.5	0.45	0.4	0.35	0.3	0.25	0.2
2500	-2.59	-2.49	-2.61	-2.14	0.79	15.36	16.68	16.42														
2000		0.69	0.73	-2.68	-2.78	-3.02	-2.92	-2.39	-1.80	13.20	8.23	6.18	5.70	5.31	5.27	5.22						
1500						6.57	6.55	6.52	6.49	6.43	6.39	6.36	6.34	6.76	6.54	6.45	6.45	-1.21	-1.21	-1.21		
1000												8.21	8.22	8.21	8.20	8.21	8.23	8.23	8.26	8.23	8.25	8.28

EGR Valve

A value highlighted in red indicates a misfire.

Table 11. EGR Valve analysis NOx emissions

rpm \ deg	90	80	70	60	50	40	30	20	10	0
5500	784.7379	879.8	702.5736	844.989	770.7371	865.2807	897.442	845.5159	809.0189	805.285
5250	894.7788	886.5188	895.3427	883.4406	671.1071	881.1649	681.4924	880.6569	804.1659	758.9216
5000	779.0965	779.7482	923.3759	817.2419	736.7613	928.2553	744.9147	767.742	743.638	788.4029
4750	703.8689	856.4089	718.0325	899.6037	772.5358	843.7577	772.9739	824.3648	792.5062	805.3245
4500	839.3511	788.1039	844.8396	798.8429	813.1813	783.5396	660.6142	740.3445	780.8648	777.5043
4250	780.9982	780.7417	780.8098	781.725	780.8586	781.7909	781.0768	780.5058	774.5963	769.7036
4000	786.2805	786.7533	682.3399	768.3901	869.3316	681.5403	774.472	776.1542	707.0011	621.8033
3750	702.0481	769.0693	768.5379	768.6908	768.467	857.8433	748.4445	821.0227	768.6181	594.0246
3500	857.3009	857.3979	858.0335	856.8544	792.8837	790.2291	923.754	920.3077	823.966	1.533086
3250	776.117	776.417	775.7455	777.9122	674.1661	947.7955	808.9283	712.079	757.8078	548.1296
3000	756.0031	756.3209	756.2294	757.2835	705.5095	842.9791	783.8933	763.0527	764.6049	509.2846
2750	735.8893	736.2248	736.7133	736.6548	736.853	738.1628	758.8583	814.0263	771.402	470.4396
2500	819.8938	819.273	819.882	818.8644	817.1692	815.8676	808.1037	796.0386	803.3257	303.5902
2250	796.6814	796.9295	796.7963	797.1521	797.0808	797.5651	798.9919	804.5543	821.0979	62.861
2000	673.6514	673.3741	674.2081	674.5634	675.4519	678.2179	683.8	701.0877	265.0805	258.92
1750	430.4591	430.5974	431.0084	433.2106	435.3701	439.4584	451.9264	493.9729	557.4617	183.7815
1500	283.9911	285.2178	283.8604	292.1187	296.284	311.2186	352.367	539.4586	118.6708	88.19873
1000	585.0678	585.1512	584.6569	584.4862	584.5742	584.7394	583.4168	580.0696	434.5468	145.0366

Table 12. EGR Valve analysis peak in-cylinder temperature

rpm \ deg	90	80	70	60	50	40	30	20	10	0
5500	2982.276	2910.856	3012.403	2892.957	2979.326	2917.78	2945.954	2932.666	2989.682	3016.382
5250	2907.981	2926.14	2907.388	2930.301	3025.746	2935.764	3023.612	2941.31	3000.076	2643.492
5000	2976.984	2984.641	2868.187	2951.005	2994.675	2872.013	3003.225	3007.487	3026.593	3042.143
4750	3012.209	2947.822	3008.179	2943.047	2984.69	2961.489	2967.225	2973.039	3000.23	3017.283
4500	2934.286	2965.291	2932.305	2951.633	2949.471	2968.623	3025.541	2991.055	2989.105	3040.118
4250	2954.338	2954.008	2955.207	2953.598	2956.508	2957.122	2962.572	2974.256	2998.732	3043.812
4000	3005.205	2998.16	3016.454	3010.028	2918.065	3028.976	2974.566	2987.282	2663.049	2332.419
3750	3034.673	3022.253	3025.614	3026.349	3026.098	2939.56	2995.255	2937.068	3031.536	2335.863
3500	3007.524	3008.161	3007.319	3007.935	2988.843	2991.034	2955.792	2976.116	3028.31	940.4045
3250	2974.268	2979.145	2975.903	2974.476	2988.234	2823.773	2943.776	3048.563	2999.112	2344.253
3000	3038.23	3038.205	3038.474	3038.633	3044.857	2961.738	3018.835	3049.425	3048.653	2371.696
2750	3102.192	3097.265	3101.046	3102.789	3101.479	3099.703	3093.894	3050.287	3098.194	2399.139
2500	3074.31	3076.391	3073.484	3075.52	3073.891	3077.605	3105.996	3100.544	3102.377	2041.13
2250	3121.415	3122.029	3121.591	3120.445	3120.739	3120.936	3119.371	3117.353	3107.046	1294.002
2000	3185.24	3185.695	3184.814	3185.355	3183.277	3181.362	3179.377	3168.227	2037.18	2034.156
1750	3293.683	3293.414	3291.004	3293.309	3292.697	3290.191	3285.596	3268.561	2796.309	2052.327
1500	3401.669	3399.491	3402.104	3400.354	3402.899	3403.695	3404.305	3408.684	2528.388	2086.146
1000	2407.53	2407.432	2409.448	2410.213	2412.842	2416.331	2425.972	2458.078	2354.262	1844.594

EGR intercooler

A value highlighted in red indicates a misfire.

Table 13. EGR Intercooler bypass analysis NOx emissions

rpm \ mm	50	45	40	35	30	25	20	15	10	5	0
5500	772.0076	708.8233	899.1448	785.4892	748.6271	712.3988	772.0163	798.4669	755.8376	806.8196	807.6556
4000	766.0475	767.6094	753.0024	792.1891	780.2411	777.8549	776.7295	773.4656	704.9018	618.7949	620.3336
3500	706.6393	706.7408	757.1997	753.91	694.26	742.89	687.411	764.6951	759.7605	753.9653	753.3403
3000	737.0367	750.1144	751.261	742.3085	770.9183	765.4964	797.9183	764.0397	760.4203	757.4364	626.5247
2500	818.8256	818.8811	818.0965	818.0627	807.7979	826.0412	794.0294	803.4408	801.0162	152.6379	152.1049
2000	674.5634	675.0235	674.6444	675.6941	678.8606	684.6538	698.2525	727.4932	262.4537	248.3466	254.7423
1000	585.2284	585.7861	584.4862	585.1918	584.2567	582.6824	577.9251	446.6366	140.5105	148.8049	151.8225

Table 14. EGR Intercooler bypass analysis peak in-cylinder temperatures

rpm \ mm	50	45	40	35	30	25	20	15	10	5	0
5500	2986.29	3001.119	2941.902	2980.83	2997.121	3009.102	2993.989	2983.207	2994.614	2990.105	2992.987
4000	3006.948	3007.686	2991.594	2969.843	2970.555	2971.975	2976.887	2985.159	2634.5	2286.221	2284.183
3500	3057.05	3057.156	3038.173	3043.279	3049.485	3014.853	3039.391	3019.592	3026.201	3019.686	3013.547
3000	3071.013	3070.954	3069.615	3077.141	3070.491	3047.266	3033.467	3061.07	3069.021	3078.519	2726.486
2500	3076.647	3076.111	3065.955	3077.674	3082.949	3077.745	3091.032	3092.771	3099.619	1677.273	1674.53
2000	3185.355	3182.868	3183.658	3184.207	3180.931	3178.799	3171.262	3156.091	2037.821	2036.931	2033.654
1000	2409.072	2409.746	2410.213	2411.904	2415.805	2425.023	2437.057	2285.93	1845.518	1830.263	1831.524

VVL/VVT

A value highlighted in red indicates a misfire.

Table 15. Variable Valve Lift analysis results

1000 rpm	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13	Case 14	Case 15	Case 16	Case 17	Case 18	Case 19	Case 20
Intake valve lift (mm)	1.11	1.11	1.11	2.22	2.22	2.22	3.33	3.33	3.33	4.44	4.44	4.44	4.44	4.44	5.55	5.55	5.55	5.55	5.55	6.66
Exhaust valve lift (mm)	1.23	2.46	3.69	1.23	2.46	3.69	1.23	2.46	3.69	6.14	7.37	8.60	9.83	11.06	6.14	7.37	8.60	9.83	11.06	6.14
NOx emissions (ppm)	617.89	0.06	0.01	605.85	632.05	658.60	601.99	614.05	636.27	653.66	666.00	675.90	684.66	691.76	640.42	651.28	660.01	666.92	672.91	631.63
Volumetric efficiency (fraction)	0.59	0.41	0.39	0.81	0.85	0.86	0.85	0.94	0.94	0.91	0.89	0.89	0.88	0.88	0.90	0.88	0.88	0.88	0.88	0.90
IMEP (bar)	4.79	-1.46	-1.20	5.71	8.45	8.82	5.99	9.65	10.07	9.96	9.81	9.72	9.67	9.61	10.05	9.82	9.83	9.79	9.72	10.12

1000 rpm	Case 21	Case 22	Case 23	Case 24	Case 25	Case 26	Case 27	Case 28	Case 29	Case 30	Case 31	Case 32	Case 33	Case 34	Case 35	Case 36	Case 37	Case 38	Case 39	Case 40
Intake valve lift (mm)	6.66	6.66	6.66	6.66	7.77	7.77	7.77	7.77	7.77	8.88	8.88	8.88	8.88	8.88	9.99	9.99	9.99	9.99	9.99	9.99
Exhaust valve lift (mm)	7.37	8.60	9.83	11.06	6.14	7.37	8.60	9.83	11.06	6.14	7.37	8.60	9.83	11.06	6.14	7.37	8.60	9.83	11.06	14.75
NOx emissions (ppm)	641.19	648.88	653.80	659.07	625.55	632.41	639.36	645.05	505.38	619.92	479.67	486.05	492.23	496.31	466.95	474.40	480.91	484.82	488.60	497.46
Volumetric efficiency (fraction)	0.88	0.88	0.87	0.87	0.89	0.87	0.87	0.87	0.83	0.89	0.83	0.83	0.82	0.82	0.84	0.83	0.82	0.82	0.81	0.81
IMEP (bar)	9.95	9.88	9.79	9.76	10.06	9.93	9.84	9.78	9.54	10.08	9.60	9.66	9.58	9.47	9.63	9.64	9.51	9.51	9.54	9.37

1000 rpm	Case 41	Case 42	Case 43	Case 44	Case 45	Case 46	Case 47	Case 48	Case 49	Case 50	Case 51	Case 52	Case 53	Case 54	Case 55	Case 56	Case 57	Case 58	Case 59	Case 60
Intake valve lift (mm)	9.99	9.99	9.99	9.99	11.10	11.10	11.10	11.10	11.10	11.10	11.10	12.22	12.22	12.22	12.22	12.22	12.22	12.22	13.33	13.33
Exhaust valve lift (mm)	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52
NOx emissions (ppm)	499.28	500.81	500.75	501.52	484.98	488.09	490.76	491.53	492.69	493.38	493.93	479.24	481.60	483.22	484.60	485.34	486.06	487.51	473.84	476.94
Volumetric efficiency (fraction)	0.80	0.80	0.80	0.80	0.81	0.80	0.80	0.80	0.80	0.80	0.79	0.80	0.80	0.80	0.79	0.79	0.79	0.79	0.80	0.79
IMEP (bar)	9.35	9.38	9.30	9.32	9.44	9.37	9.39	9.36	9.35	9.31	9.26	9.37	9.37	9.37	9.30	9.30	9.27	9.25	9.34	9.34

1000 rpm	Case 61	Case 62	Case 63	Case 64	Case 65	Case 66	Case 67	Case 68	Case 69	Case 70	Case 71	Case 72	Case 73	Case 74	Case 75	Case 76	Case 77	Case 78	Case 79
Intake valve lift (mm)	13.33	13.33	13.33	13.33	13.33	14.44	14.44	14.44	14.44	14.44	14.44	14.44	15.55	15.55	15.55	15.55	15.55	15.55	15.55
Exhaust valve lift (mm)	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66
NOx emissions (ppm)	477.59	477.86	480.15	480.35	481.68	469.87	471.71	471.97	473.41	473.64	475.40	475.37	465.61	466.83	468.35	469.36	469.97	470.72	175.94
Volumetric efficiency (fraction)	0.79	0.79	0.79	0.79	0.78	0.79	0.79	0.79	0.78	0.78	0.78	0.78	0.79	0.79	0.78	0.78	0.78	0.78	0.69
IMEP (bar)	9.32	9.28	9.27	9.25	9.23	9.32	9.30	9.34	9.26	9.22	9.22	9.19	9.30	9.29	9.30	9.25	9.20	9.18	-0.74

3000 rpm	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13	Case 14	Case 15	Case 16	Case 17	Case 18	Case 19	Case 20
Intake valve lift (mm)	6.66	6.66	6.66	6.66	7.77	7.77	7.77	7.77	8.88	8.88	8.88	8.88	9.99	9.99	9.99	9.99	11.10	11.10	11.10	11.10
Exhaust valve lift (mm)	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29
Nox emissions (ppm)	668.87	550.49	182.55	178.89	756.46	769.03	782.84	795.39	742.43	749.69	763.17	772.32	732.13	740.51	752.10	776.62	697.02	764.44	767.69	769.46
Volumetric efficiency (fraction)	1.25	1.24	1.05	1.03	1.35	1.35	1.34	1.34	1.35	1.35	1.36	1.36	1.36	1.36	1.36	1.37	1.36	1.40	1.40	1.40
IMEP (bar)	16.32	15.79	12.91	12.58	17.81	17.94	17.95	17.95	17.74	17.91	18.06	18.10	17.75	17.97	18.02	18.38	17.90	18.67	18.84	18.96

3000 rpm	Case 21	Case 22	Case 23	Case 24	Case 25	Case 26	Case 27	Case 28	Case 29	Case 30	Case 31	Case 32	Case 33	Case 34	Case 35	Case 36	Case 37	Case 38	Case 39	Case 40
Intake valve lift (mm)	11.10	11.10	11.10	11.10	12.22	12.22	12.22	12.22	13.33	13.33	13.33	13.33	13.33	14.44	14.44	14.44	14.44	15.55	15.55	15.55
Exhaust valve lift (mm)	13.52	14.75	15.98	19.66	12.29	13.52	14.75	15.98	12.29	13.52	14.75	15.98	19.66	12.29	13.52	14.75	15.98	12.29	15.98	19.66
Nox emissions (ppm)	771.64	775.11	752.71	744.11	767.61	769.26	770.72	772.97	766.39	767.49	769.15	769.93	773.09	765.32	766.39	768.30	769.47	764.59	768.47	773.46
Volumetric efficiency (fraction)	1.40	1.38	1.40	1.37	1.41	1.41	1.41	1.38	1.41	1.41	1.41	1.41	1.38	1.41	1.41	1.41	1.41	1.41	1.41	1.41
IMEP (bar)	19.06	18.53	19.18	19.01	18.96	19.06	19.12	18.55	18.96	19.08	19.13	19.18	18.62	18.95	19.05	19.12	19.19	18.93	19.17	19.23

5500 rpm	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	Case 11	Case 12	Case 13	Case 14	Case 15	Case 16	Case 17	Case 18	Case 19	Case 20
Intake valve lift (mm)	6.66	6.66	6.66	6.66	7.77	7.77	7.77	7.77	8.88	8.88	8.88	8.88	9.99	9.99	9.99	9.99	11.10	11.10	11.10	11.10
Exhaust valve lift (mm)	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29	8.60	9.83	11.06	12.29
Nox emissions (ppm)	786.36	706.27	780.66	675.72	735.49	787.84	851.27	657.61	836.63	754.49	895.58	872.67	634.41	842.54	826.13	859.48	775.12	800.07	715.41	730.50
Volumetric efficiency (fraction)	0.55	0.61	0.54	0.57	0.63	0.65	0.64	0.52	0.59	0.62	0.61	0.63	0.61	0.64	0.64	0.66	0.63	0.66	0.67	0.68
IMEP (bar)	5.07	6.23	5.03	5.81	6.10	6.43	6.43	-3.11	4.94	6.17	5.34	6.14	6.29	6.06	6.45	6.54	5.91	6.54	7.04	7.27

5500 rpm	Case 21	Case 22	Case 23	Case 24	Case 25	Case 26	Case 27	Case 28	Case 29	Case 30	Case 31	Case 32	Case 33	Case 34	Case 35	Case 36	Case 37	Case 38	Case 39	Case 40
Intake valve lift (mm)	11.10	11.10	11.10	11.10	11.10	11.10	11.10	12.22	12.22	12.22	12.22	12.22	12.22	12.22	13.33	13.33	13.33	13.33	13.33	13.33
Exhaust valve lift (mm)	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43
Nox emissions (ppm)	730.50	890.55	629.83	776.75	780.17	777.47	335.56	885.63	590.22	734.17	805.15	780.72	792.43	790.13	856.87	568.19	951.84	735.56	776.24	785.53
Volumetric efficiency (fraction)	0.68	0.67	0.69	0.70	0.70	0.69	0.56	0.67	0.68	0.70	0.69	0.71	0.69	0.72	0.69	0.69	0.69	0.72	0.70	0.71
IMEP (bar)	7.27	6.25	8.10	7.72	7.77	7.39	-3.94	6.15	7.41	7.68	7.42	7.92	7.64	8.03	6.94	7.63	6.25	7.87	7.63	7.93

5500 rpm	Case 41	Case 42	Case 43	Case 44	Case 45	Case 46	Case 47	Case 48	Case 49	Case 50	Case 51	Case 52	Case 53	Case 54	Case 55
Intake valve lift (mm)	13.33	14.44	14.44	14.44	14.44	14.44	14.44	14.44	15.55	15.55	15.55	15.55	15.55	15.55	15.55
Exhaust valve lift (mm)	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66	12.29	13.52	14.75	15.98	17.20	18.43	19.66
Nox emissions (ppm)	784.78	684.53	872.87	735.35	886.82	897.22	827.75	799.30	859.92	676.92	777.05	889.51	681.08	826.39	834.69
Volumetric efficiency (fraction)	0.72	0.70	0.70	0.72	0.70	0.71	0.69	0.69	0.71	0.71	0.70	0.71	0.72	0.70	0.70
IMEP (bar)	7.93	7.62	6.76	7.70	6.79	7.04	6.58	6.82	6.99	7.88	6.52	6.97	8.24	6.52	6.55

DOE correlation

Attached are the Morris sensitivity analysis correlation coefficients results obtained for each engine speed studied.

Table 16. DOE Morris sensitivity analysis correlation coefficients

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	1000rpm	Boost Pressure	Factor	1											
2	1000rpm	EGR Valve	Factor	0.0396	1										
3	1000rpm	EGR Bypass	Factor	-0.0324	0.0803	1									
4	1000rpm	Intake Valve Lift	Factor	-0.0254	0.0777	-0.0109	1								
5	1000rpm	Exhaust Valve Lift	Factor	0.0275	0.1021	-0.0168	-0.0325	1							
6	1000rpm	Brake Torque	Response	-0.0415	-0.5049	-0.5615	0.1757	-0.5987	1						
7	1000rpm	Brake Power	Response	-0.0415	-0.5049	-0.5615	0.1757	-0.5987	1	1					
8	1000rpm	Volumetric Efficiency	Response	-0.0341	-0.5426	-0.5054	-0.0624	-0.6392	0.9601	0.9601	1				
9	1000rpm	EGR %	Response	0.0293	0.6851	0.5372	-0.1161	0.4310	-0.9453	-0.9453	-0.9259	1			
10	1000rpm	AFR	Response	-0.0384	-0.1429	-0.1128	0.6262	-0.7591	0.6946	0.6946	0.5751	-0.5493	1		
11	1000rpm	Cylinder 1 NOx at EVO	Response	0.0378	0.2129	0.1826	-0.5397	0.7400	-0.7311	-0.7311	-0.6135	0.5988	-0.9577	1	
12	1000rpm	Cylinder 1 Max. Temp.	Response	-0.0348	-0.5882	-0.3618	0.0395	-0.7044	0.9477	0.9477	0.9682	-0.9229	0.6764	-0.7126	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	1500rpm	Boost Pressure	Factor	1											
2	1500rpm	EGR Valve	Factor	-0.0719	1										
3	1500rpm	EGR Bypass	Factor	-0.0666	0.0028	1									
4	1500rpm	Intake Valve Lift	Factor	0.0624	-0.0475	0.1172	1								
5	1500rpm	Exhaust Valve Lift	Factor	-0.0064	0.0359	-0.0901	-0.0538	1							
6	1500rpm	Brake Torque	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1						
7	1500rpm	Brake Power	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1	1					
8	1500rpm	Volumetric Efficiency	Response	0.1114	0.0031	0.0115	0.5364	0.2887	0.9454	0.9454	1				
9	1500rpm	EGR %	Response	-0.3498	0.3029	0.1725	-0.4421	0.3802	-0.5778	-0.5778	-0.4400	1			
10	1500rpm	AFR	Response	-0.2296	-0.0519	-0.1271	-0.2158	0.5259	-0.3563	-0.3563	-0.2619	0.7732	1		
11	1500rpm	Cylinder 1 NOx at EVO	Response	-0.3461	0.2225	0.1295	-0.3969	0.3463	-0.6708	-0.6708	-0.6223	0.8632	0.6814	1	
12	1500rpm	Cylinder 1 Max. Temp.	Response	-0.2598	0.0422	-0.0593	-0.2059	0.6277	-0.2197	-0.2197	-0.1039	0.7763	0.9352	0.6662	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	2000rpm	Boost Pressure	Factor	1											
2	2000rpm	EGR Valve	Factor	-0.0719	1										
3	2000rpm	EGR Bypass	Factor	-0.0666	0.0028	1									
4	2000rpm	Intake Valve Lift	Factor	0.0624	-0.0475	0.1172	1								
5	2000rpm	Exhaust Valve Lift	Factor	-0.0064	0.0359	-0.0901	-0.0538	1							
6	2000rpm	Brake Torque	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1						
7	2000rpm	Brake Power	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1	1					
8	2000rpm	Volumetric Efficiency	Response	0.1114	0.0031	0.0115	0.5364	0.2887	0.9454	0.9454	1				
9	2000rpm	EGR %	Response	-0.3498	0.3029	0.1725	-0.4421	0.3802	-0.5778	-0.5778	-0.4400	1			
10	2000rpm	AFR	Response	-0.2296	-0.0519	-0.1271	-0.2158	0.5259	-0.3563	-0.3563	-0.2619	0.7732	1		
11	2000rpm	Cylinder 1 NOx at EVO	Response	-0.3461	0.2225	0.1295	-0.3969	0.3463	-0.6708	-0.6708	-0.6223	0.8632	0.6814	1	
12	2000rpm	Cylinder 1 Max. Temp.	Response	-0.2598	0.0422	-0.0593	-0.2059	0.6277	-0.2197	-0.2197	-0.1039	0.7763	0.9352	0.6662	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	2500rpm	Boost Pressure	Factor	1											
2	2500rpm	EGR Valve	Factor	-0.0719	1										
3	2500rpm	EGR Bypass	Factor	-0.0666	0.0028	1									
4	2500rpm	Intake Valve Lift	Factor	0.0624	-0.0475	0.1172	1								
5	2500rpm	Exhaust Valve Lift	Factor	-0.0064	0.0359	-0.0901	-0.0538	1							
6	2500rpm	Brake Torque	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1						
7	2500rpm	Brake Power	Response	0.1994	-0.0802	-0.0519	0.5108	0.2930	1	1					
8	2500rpm	Volumetric Efficiency	Response	0.1114	0.0031	0.0115	0.5364	0.2887	0.9454	0.9454	1				
9	2500rpm	EGR %	Response	-0.3498	0.3029	0.1725	-0.4421	0.3802	-0.5778	-0.5778	-0.4400	1			
10	2500rpm	AFR	Response	-0.2296	-0.0519	-0.1271	-0.2158	0.5259	-0.3563	-0.3563	-0.2619	0.7732	1		
11	2500rpm	Cylinder 1 NOx at EVO	Response	-0.3461	0.2225	0.1295	-0.3969	0.3463	-0.6708	-0.6708	-0.6223	0.8632	0.6814	1	
12	2500rpm	Cylinder 1 Max. Temp.	Response	-0.2598	0.0422	-0.0593	-0.2059	0.6277	-0.2197	-0.2197	-0.1039	0.7763	0.9352	0.6662	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	3000rpm	Boost Pressure	Factor	1											
2	3000rpm	EGR Valve	Factor	-0.0168	1										
3	3000rpm	EGR Bypass	Factor	0.0149	0.0498	1									
4	3000rpm	Intake Valve Lift	Factor	0.0210	0.0645	-0.0342	1								
5	3000rpm	Exhaust Valve Lift	Factor	0.0623	-0.0008	-0.0324	0.0443	1							
6	3000rpm	Brake Torque	Response	-0.1950	-0.4794	-0.3862	-0.2225	0.1872	1						
7	3000rpm	Brake Power	Response	-0.1950	-0.4794	-0.3862	-0.2225	0.1872	1	1					
8	3000rpm	Volumetric Efficiency	Response	-0.1399	-0.4294	-0.4539	0.3818	-0.0520	0.4265	0.4265	1				
9	3000rpm	EGR %	Response	0.1507	0.4673	0.2837	0.4674	0.0799	-0.8935	-0.8935	-0.2893	1			
10	3000rpm	AFR	Response	0.1085	0.1596	-0.0143	0.0849	0.0665	-0.2692	-0.2692	0.3277	0.2166	1		
11	3000rpm	Cylinder 1 NOx at EVO	Response	-0.0397	-0.2058	-0.0362	-0.1906	0.0826	0.5158	0.5158	-0.3128	-0.4656	-0.9013	1	
12	3000rpm	Cylinder 1 Max. Temp.	Response	-0.0804	-0.2323	-0.2049	-0.2680	-0.1638	0.5132	0.5132	0.4533	-0.6646	0.4511	-0.2150	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	3500rpm	Boost Pressure	Factor	1											
2	3500rpm	EGR Valve	Factor	-0.0168	1										
3	3500rpm	EGR Bypass	Factor	0.0149	0.0498	1									
4	3500rpm	Intake Valve Lift	Factor	0.0210	0.0645	-0.0342	1								
5	3500rpm	Exhaust Valve Lift	Factor	0.0623	-0.0008	-0.0324	0.0443	1							
6	3500rpm	Brake Torque	Response	-0.1950	-0.4794	-0.3862	-0.2225	0.1872	1						
7	3500rpm	Brake Power	Response	-0.1950	-0.4794	-0.3862	-0.2225	0.1872	1	1					
8	3500rpm	Volumetric Efficiency	Response	-0.1399	-0.4294	-0.4539	0.3818	-0.0520	0.4265	0.4265	1				
9	3500rpm	EGR %	Response	0.1507	0.4673	0.2837	0.4674	0.0799	-0.8935	-0.8935	-0.2893	1			
10	3500rpm	AFR	Response	0.1085	0.1596	-0.0143	0.0849	0.0665	-0.2692	-0.2692	0.3277	0.2166	1		
11	3500rpm	Cylinder 1 NOx at EVO	Response	-0.0397	-0.2058	-0.0362	-0.1906	0.0826	0.5158	0.5158	-0.3128	-0.4656	-0.9013	1	
12	3500rpm	Cylinder 1 Max. Temp.	Response	-0.0804	-0.2323	-0.2049	-0.2680	-0.1638	0.5132	0.5132	0.4533	-0.6646	0.4511	-0.2150	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	4000rpm	Boost Pressure	Factor	1											
2	4000rpm	EGR Valve	Factor	-0.0160	1										
3	4000rpm	EGR Bypass	Factor	-0.0376	-0.1056	1									
4	4000rpm	Intake Valve Lift	Factor	0.0774	-0.0499	-0.0071	1								
5	4000rpm	Exhaust Valve Lift	Factor	0.0229	-0.0249	-0.0344	-0.0429	1							
6	4000rpm	Brake Torque	Response	-0.0835	0.0682	-0.0428	-0.6031	0.5954	1						
7	4000rpm	Brake Power	Response	-0.0835	0.0682	-0.0428	-0.6031	0.5954	1	1					
8	4000rpm	Volumetric Efficiency	Response	-0.2388	-0.3558	-0.2618	0.5851	-0.0938	-0.2589	-0.2589	1				
9	4000rpm	EGR %	Response	0.0735	0.5019	0.2476	0.6263	0.0242	-0.3801	-0.3801	-0.0623	1			
10	4000rpm	AFR	Response	-0.1072	-0.1202	-0.1462	0.0324	-0.0327	-0.1835	-0.1835	0.3124	-0.1883	1		
11	4000rpm	Cylinder 1 NOx at EVO	Response	-0.0217	0.1902	0.2342	0.4010	0.0459	-0.0712	-0.0712	0.0129	0.5815	-0.7972	1	
12	4000rpm	Cylinder 1 Max. Temp.	Response	0.1170	-0.2598	-0.0905	-0.5602	0.1345	0.3269	0.3269	-0.2328	-0.6635	0.5596	-0.8853	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	4500rpm	Boost Pressure	Factor	1											
2	4500rpm	EGR Valve	Factor	0.0484	1										
3	4500rpm	EGR Bypass	Factor	0.0480	0.1495	1									
4	4500rpm	Intake Valve Lift	Factor	0.0430	0.1032	0.0060	1								
5	4500rpm	Exhaust Valve Lift	Factor	-0.0724	0.0247	-0.0567	-0.0932	1							
6	4500rpm	Brake Torque	Response	0.2744	0.1383	0.0794	-0.2704	0.1869	1						
7	4500rpm	Brake Power	Response	0.2744	0.1383	0.0794	-0.2704	0.1869	1	1					
8	4500rpm	Volumetric Efficiency	Response	0.4011	-0.2193	-0.3035	0.3321	0.0815	0.3010	0.3010	1				
9	4500rpm	EGR %	Response	-0.0917	0.5150	0.1760	0.5204	0.1434	-0.4575	-0.4575	-0.1815	1			
10	4500rpm	AFR	Response	-0.1655	0.0003	0.0309	0.0755	0.0018	-0.4446	-0.4446	0.0069	0.2488	1		
11	4500rpm	Cylinder 1 NOx at EVO	Response	0.2368	0.0921	0.0403	-0.0330	0.0844	0.3018	0.3018	-0.0348	-0.1464	-0.9008	1	
12	4500rpm	Cylinder 1 Max. Temp.	Response	-0.1943	-0.1779	-0.0370	-0.0758	-0.1032	-0.0346	-0.0346	0.0849	-0.1230	0.7586	-0.9255	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	5000rpm	Boost Pressure	Factor	1											
2	5000rpm	EGR Valve	Factor	0.0573	1										
3	5000rpm	EGR Bypass	Factor	0.0667	-0.0144	1									
4	5000rpm	Intake Valve Lift	Factor	0.0339	-0.0194	-0.0977	1								
5	5000rpm	Exhaust Valve Lift	Factor	-0.0885	0.0512	0.0851	-0.0846	1							
6	5000rpm	Brake Torque	Response	0.0142	-0.0649	0.0439	-0.2632	0.1083	1						
7	5000rpm	Brake Power	Response	0.0142	-0.0649	0.0439	-0.2632	0.1083	1	1					
8	5000rpm	Volumetric Efficiency	Response	0.1000	-0.3456	-0.3479	0.0811	0.0163	0.3108	0.3108	1				
9	5000rpm	EGR %	Response	0.0519	0.2560	0.1756	0.3904	0.3813	-0.4032	-0.4032	-0.0825	1			
10	5000rpm	AFR	Response	0.0253	0.0535	-0.0650	0.0566	-0.1092	-0.0649	-0.0649	0.0028	-0.0381	1		
11	5000rpm	Cylinder 1 NOx at EVO	Response	-0.0174	0.0197	0.0328	-0.0528	0.1385	-0.3501	-0.3501	-0.1159	0.0089	-0.7821	1	
12	5000rpm	Cylinder 1 Max. Temp.	Response	0.0059	-0.1266	-0.0389	-0.0245	-0.1380	0.6199	0.6199	0.2113	-0.2846	0.6040	-0.8928	1

	Dataset	Variable	Type	Factors					Responses						
Type				Boost Pressure	EGR Valve	EGR Bypass	Intake Valve Lift	Exhaust Valve Lift	Brake Torque	Brake Power	Volumetric Efficiency	EGR %	AFR	Cylinder 1 NOx at EVO	Cylinder 1 Max. Temp.
1	5500rpm	Boost Pressure	Factor	1											
2	5500rpm	EGR Valve	Factor	-0.1311	1										
3	5500rpm	EGR Bypass	Factor	-0.0571	0.0321	1									
4	5500rpm	Intake Valve Lift	Factor	0.0632	0.0299	0.0329	1								
5	5500rpm	Exhaust Valve Lift	Factor	0.0375	-0.0288	0.0845	-0.0032	1							
6	5500rpm	Brake Torque	Response	0.0099	-0.0842	-0.1104	0.0148	0.0336	1						
7	5500rpm	Brake Power	Response	0.0099	-0.0842	-0.1104	0.0148	0.0336	1	1					
8	5500rpm	Volumetric Efficiency	Response	0.0818	-0.3067	-0.2689	0.2680	0.1270	0.6740	0.6740	1				
9	5500rpm	EGR %	Response	0.0018	0.1592	0.1457	0.1897	0.2698	-0.7338	-0.7338	-0.5494	1			
10	5500rpm	AFR	Response	-0.0263	0.0252	0.0804	-0.0662	0.0450	-0.4713	-0.4713	-0.1750	0.0878	1		
11	5500rpm	Cylinder 1 NOx at EVO	Response	0.0209	0.0505	0.0458	0.1618	0.0847	-0.3773	-0.3773	-0.1676	0.3703	-0.2734	1	
12	5500rpm	Cylinder 1 Max. Temp.	Response	0.0056	-0.1093	-0.0620	-0.0309	-0.0770	0.7871	0.7871	0.4777	-0.6096	-0.1068	-0.7768	1

Metamodels RMS Error

Table 17. RMS Error for all initial metamodels studied

rpm / Metamodel	1000	1250	1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500
Brake_Torque_Kriging	3.975	225.483	277.106	130.682	5.458	5.816	11.982	1.959	1.538	1.607	0.672	6.443	1.919	1.520	2.174	1.839	2.461	2.136	1.829
Brake_Torque_Linear	28.179	279.403	380.327	257.156	5.888	5.315	12.961	5.946	6.361	7.346	8.411	10.389	8.360	7.072	6.774	6.016	7.857	8.241	6.046
Brake_Torque_Quadratic	11.737	224.079	302.270	194.576	5.330	5.174	11.992	1.373	1.963	2.121	2.095	7.270	2.648	2.438	2.418	2.211	4.547	4.234	2.538
Brake_Torque_Cubic	3.520	232.158	296.435	196.181	5.284	5.200	11.915	1.172	1.192	1.266	0.711	7.131	1.349	1.239	1.468	1.378	3.842	3.609	1.482
Brake_Torque_MLP	1.727	183.989	214.587	33.415	4.832	4.914	11.507	1.055	1.100	1.034	0.348	4.049	1.050	1.103	1.132	1.075	1.262	1.233	1.051
Brake_Torque_IDW	25.252	231.749	283.979	125.270	5.674	5.463	11.707	8.016	8.384	9.273	8.808	11.937	9.298	10.038	9.390	7.862	11.036	9.917	7.921
Brake_Power_Kriging	0.401	39.603	57.313	32.024	1.529	1.820	4.203	0.655	0.676	0.842	0.358	3.231	1.298	0.891	1.402	1.441	1.829	1.596	1.801
Brake_Power_Linear	3.957	49.046	80.115	63.197	1.654	1.680	4.550	2.296	2.680	3.353	4.134	5.471	4.696	4.221	4.281	4.013	5.517	6.076	4.670
Brake_Power_Quadratic	1.650	39.334	63.692	47.818	1.497	1.635	4.211	0.530	0.826	0.968	1.030	3.828	1.487	1.455	1.527	1.475	3.202	3.122	1.962
Brake_Power_Cubic	0.487	39.769	62.148	48.131	1.481	1.645	4.187	0.452	0.502	0.575	0.349	3.756	0.755	0.739	0.926	0.919	2.739	2.661	1.135
Brake_Power_MLP	0.243	33.693	45.244	23.347	1.364	1.622	4.025	0.398	0.476	0.476	0.168	2.077	0.609	0.667	0.702	0.748	0.924	1.114	0.789
Brake_Power_IDW	3.546	40.681	59.819	30.786	1.594	1.726	4.110	3.096	3.532	4.232	4.329	6.286	5.223	5.991	5.934	5.245	7.749	7.311	6.118
Volumentric_efficiency_Kriging	0.002	0.134	0.189	0.098	0.001	0.001	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.001	0.000
Volumentric_efficiency_Linear	0.030	0.164	0.264	0.189	0.001	0.001	0.009	0.002	0.004	0.004	0.005	0.006	0.006	0.005	0.005	0.005	0.005	0.005	0.004
Volumentric_efficiency_Quadratic	0.013	0.132	0.209	0.144	0.001	0.000	0.008	0.001	0.001	0.001	0.002	0.003	0.002	0.002	0.002	0.002	0.002	0.002	0.001
Volumentric_efficiency_Cubic	0.003	0.132	0.204	0.145	0.001	0.000	0.008	0.000	0.000	0.001	0.001	0.003	0.001	0.001	0.001	0.001	0.002	0.002	0.001
Volumentric_efficiency_MLP	0.001	0.106	0.156	0.005	0.000	0.000	0.006	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumentric_efficiency_IDW	0.028	0.138	0.198	0.092	0.002	0.001	0.008	0.003	0.004	0.005	0.006	0.006	0.006	0.007	0.007	0.006	0.008	0.006	0.006
EGR_Percentage_Kriging	0.038	6.135	7.328	3.545	0.092	0.131	0.322	0.008	0.008	0.008	0.018	0.061	0.017	0.019	0.019	0.018	0.015	0.029	0.020
EGR_Percentage_Linear	0.714	7.660	10.199	7.139	0.092	0.093	0.349	0.096	0.148	0.183	0.229	0.232	0.283	0.288	0.286	0.312	0.347	0.333	0.269
EGR_Percentage_Quadratic	0.234	6.110	8.147	5.423	0.067	0.082	0.314	0.030	0.051	0.059	0.075	0.090	0.099	0.099	0.094	0.097	0.122	0.124	0.099
EGR_Percentage_Cubic	0.064	6.257	8.028	5.462	0.064	0.083	0.316	0.017	0.024	0.029	0.042	0.063	0.047	0.050	0.052	0.050	0.065	0.080	0.055
EGR_Percentage_MLP	0.013	3.828	6.161	2.117	0.058	0.083	0.298	0.003	0.001	0.001	0.001	0.045	0.002	0.002	0.001	0.002	0.003	0.005	0.008
EGR_Percentage_IDW	0.658	6.336	7.649	3.489	0.148	0.113	0.300	0.131	0.164	0.185	0.201	0.243	0.275	0.236	0.306	0.315	0.291	0.367	0.251
Engine_AFR_Kriging	0.019	1.418	0.114	0.094	0.067	0.044	0.178	0.004	0.002	0.002	0.001	0.048	0.002	0.003	0.003	0.004	0.010	0.023	0.011
Engine_AFR_Linear	0.064	1.753	0.096	0.099	0.070	0.043	0.222	0.003	0.001	0.002	0.001	0.086	0.001	0.005	0.008	0.018	0.059	0.053	0.013
Engine_AFR_Quadratic	0.018	1.412	0.096	0.090	0.068	0.044	0.204	0.003	0.001	0.002	0.000	0.058	0.001	0.005	0.008	0.012	0.041	0.057	0.013
Engine_AFR_Cubic	0.017	1.419	0.099	0.088	0.068	0.044	0.204	0.003	0.001	0.002	0.000	0.058	0.002	0.005	0.008	0.012	0.042	0.059	0.013
Engine_AFR_MLP	0.011	0.960	0.100	0.076	0.060	0.042	0.148	0.002	0.001	0.002	0.000	0.035	0.001	0.002	0.001	0.002	0.006	0.007	0.004
Engine_AFR_IDW	0.100	1.455	0.099	0.091	0.068	0.043	0.169	0.003	0.002	0.002	0.001	0.057	0.001	0.004	0.006	0.011	0.040	0.046	0.013
Cyl1_NOx_at_EVO_Kriging	2.390	56.213	27.374	12.362	5.391	3.973	9.900	1.635	1.659	2.217	0.226	3.515	2.027	1.991	1.909	2.587	2.149	2.818	2.205
Cyl1_NOx_at_EVO_Linear	10.555	68.965	33.608	22.718	6.037	3.188	11.260	2.353	1.769	2.173	2.422	3.341	4.265	2.742	2.673	3.260	4.831	6.036	4.670
Cyl1_NOx_at_EVO_Quadratic	4.233	55.194	30.395	16.836	4.566	2.458	10.988	1.244	1.412	1.396	0.940	2.652	2.364	1.587	1.594	1.721	3.295	4.018	2.278
Cyl1_NOx_at_EVO_Cubic	3.639	56.210	30.111	16.844	4.407	2.481	11.020	1.230	1.331	1.419	0.764	2.608	1.949	1.434	1.341	1.677	3.247	3.961	2.026
Cyl1_NOx_at_EVO_MLP	1.545	41.441	23.443	5.255	3.782	2.327	8.736	1.174	1.276	1.402	0.139	2.262	1.204	1.037	1.115	1.147	1.525	1.533	1.147
Cyl1_NOx_at_EVO_IDW	9.547	57.087	26.626	14.670	7.313	4.299	9.247	2.446	1.496	2.936	3.660	5.180	5.038	5.038	4.071	4.153	4.092	6.136	3.863
Cyl1_Max_Temp_Kriging	3.511	135.162	124.400	62.390	6.038	10.116	9.402	4.561	5.346	5.151	3.091	6.678	3.681	2.832	3.552	3.541	2.752	4.639	5.310
Cyl1_Max_Temp_Linear	15.251	164.124	175.407	119.831	3.577	10.065	9.954	3.500	4.361	4.087	4.119	6.321	5.152	4.623	4.751	5.246	6.307	5.786	5.672
Cyl1_Max_Temp_Quadratic	5.741	131.796	138.430	91.277	3.653	10.033	9.871	3.303	3.718	2.898	2.545	5.713	2.776	2.191	2.255	2.038	3.245	3.681	4.711
Cyl1_Max_Temp_Cubic	2.375	133.229	134.978	92.010	3.508	10.126	9.870	3.282	3.364	2.970	2.576	5.708	2.572	1.834	1.960	1.621	2.642	3.478	4.431
Cyl1_Max_Temp_MLP	2.151	119.216	100.015	51.535	3.695	9.876	8.133	3.304	3.516	2.946	1.895	5.536	2.457	1.473	1.546	1.351	1.566	2.808	4.307
Cyl1_Max_Temp_IDW	15.239	136.818	131.224	59.526	5.112	9.964	8.362	3.606	4.536	4.773	3.838	6.415	5.089	3.936	4.935	4.808	5.388	5.532	5.243

Table 18. MLP metamodels RMS Error

rpm / Metamodel	1000	1250	1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500
Brake_Torque_10_5_trial1	1.669	188.136	212.304	54.660	4.830	5.070	11.243	1.041	1.158	1.045	0.340	4.283	1.074	1.128	1.141	1.112	1.288	1.183	0.949
Brake_Torque_10_5_trial2	1.571	182.364	186.748	85.139	4.778	4.966	11.347	1.000	1.111	1.031	0.348	3.848	1.038	1.132	1.147	1.131	1.293	1.209	1.068
Brake_Torque_10_5_trial3	1.672	206.436	217.989	129.792	4.903	5.002	11.519	1.045	1.133	1.038	0.315	4.215	1.084	1.112	1.140	1.086	1.243	1.378	1.001
Brake_Torque_10_5_5_trial1	1.656	140.965	216.781	92.519	4.832	4.965	11.298	1.017	1.147	1.047	0.371	4.069	1.074	1.110	1.146	1.095	1.227	1.300	1.032
Brake_Torque_10_5_5_trial2	1.658	183.371	210.402	69.723	4.944	4.761	11.667	1.016	1.137	1.019	0.379	4.003	1.059	1.098	1.114	1.103	1.237	1.405	1.040
Brake_Torque_10_5_5_trial3	1.674	160.982	210.920	53.410	5.025	4.988	11.189	1.006	1.133	1.039	0.336	3.997	1.076	1.094	1.108	1.104	1.221	1.287	1.084
Brake_Power_10_5_trial1	0.238	28.720	37.150	1.860	1.396	1.624	4.577	0.393	0.473	0.479	0.169	2.149	0.599	0.642	0.727	0.759	0.988	0.905	0.798
Brake_Power_10_5_trial2	0.224	31.543	37.976	15.159	1.300	1.484	4.063	0.398	0.485	0.468	0.181	2.069	0.602	0.641	0.731	0.716	0.928	0.987	0.815
Brake_Power_10_5_trial3	0.228	31.553	42.777	1.788	1.299	1.695	4.051	0.401	0.495	0.474	0.175	2.056	0.592	0.640	0.720	0.762	0.952	1.007	0.754
Brake_Power_10_5_5_trial1	0.217	21.215	45.346	18.444	1.313	1.595	4.060	0.398	0.480	0.468	0.186	2.121	0.606	0.654	0.717	0.742	0.892	1.049	0.864
Brake_Power_10_5_5_trial2	0.218	31.753	41.058	14.250	1.412	1.574	3.986	0.385	0.468	0.469	0.173	2.104	0.605	0.639	0.725	0.711	0.885	1.059	0.834
Brake_Power_10_5_5_trial3	0.228	30.148	43.758	4.813	1.400	1.576	4.164	0.395	0.485	0.474	0.188	2.138	0.609	0.662	0.730	0.722	0.865	0.935	0.816
Volumetric_Efficiency_10_5_trial1	0.001	0.096	0.151	0.059	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumetric_Efficiency_10_5_trial2	0.001	0.105	0.149	0.008	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumetric_Efficiency_10_5_trial3	0.001	0.112	0.156	0.051	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumetric_Efficiency_10_5_5_trial1	0.001	0.074	0.140	0.005	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumetric_Efficiency_10_5_5_trial2	0.001	0.106	0.135	0.045	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Volumetric_Efficiency_10_5_5_trial3	0.001	0.109	0.163	0.092	0.000	0.000	0.007	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EGR_Percentage_10_5_trial1	0.012	5.306	4.678	0.491	0.055	0.080	0.291	0.004	0.001	0.001	0.001	0.046	0.002	0.002	0.001	0.003	0.004	0.005	0.008
EGR_Percentage_10_5_trial2	0.013	3.838	5.675	0.183	0.056	0.083	0.301	0.004	0.001	0.001	0.001	0.043	0.002	0.002	0.001	0.002	0.005	0.005	0.008
EGR_Percentage_10_5_trial3	0.012	5.209	6.122	0.188	0.058	0.081	0.319	0.003	0.001	0.001	0.001	0.043	0.002	0.002	0.002	0.002	0.004	0.008	0.007
EGR_Percentage_10_5_5_trial1	0.012	4.508	5.377	0.402	0.056	0.078	0.295	0.003	0.001	0.001	0.001	0.045	0.002	0.002	0.002	0.002	0.003	0.005	0.007
EGR_Percentage_10_5_5_trial2	0.012	4.609	4.723	0.793	0.058	0.081	0.292	0.006	0.001	0.001	0.001	0.045	0.002	0.001	0.002	0.001	0.004	0.006	0.007
EGR_Percentage_10_5_5_trial3	0.012	4.539	6.042	0.235	0.056	0.078	0.291	0.004	0.001	0.001	0.001	0.049	0.002	0.002	0.001	0.002	0.005	0.005	0.007
AFR_10_5_trial1	0.012	0.800	0.094	0.077	0.056	0.042	0.147	0.002	0.002	0.002	0.000	0.036	0.001	0.002	0.002	0.002	0.004	0.007	0.005
AFR_10_5_trial2	0.012	1.118	0.094	0.076	0.058	0.042	0.159	0.003	0.001	0.002	0.000	0.040	0.001	0.002	0.002	0.002	0.003	0.011	0.004
AFR_10_5_trial3	0.011	1.012	0.095	0.076	0.058	0.041	0.156	0.002	0.001	0.002	0.000	0.036	0.001	0.002	0.001	0.002	0.005	0.010	0.005
AFR_10_5_5_trial1	0.011	1.158	0.092	0.078	0.056	0.042	0.167	0.002	0.001	0.002	0.000	0.036	0.001	0.002	0.002	0.002	0.003	0.010	0.005
AFR_10_5_5_trial2	0.011	1.216	0.088	0.078	0.060	0.042	0.159	0.002	0.001	0.002	0.000	0.036	0.001	0.002	0.002	0.002	0.006	0.010	0.004
AFR_10_5_5_trial3	0.011	1.181	0.091	0.075	0.057	0.040	0.168	0.002	0.001	0.002	0.000	0.039	0.001	0.002	0.001	0.002	0.005	0.009	0.004
NOx_at_EVO_10_5_trial1	1.494	36.014	24.177	3.580	3.913	2.369	8.229	1.103	1.256	1.396	0.118	2.087	1.201	1.045	1.134	1.132	1.471	1.330	1.010
NOx_at_EVO_10_5_trial2	1.479	36.774	22.939	4.820	3.789	2.342	9.651	1.168	1.252	1.367	0.124	2.179	1.188	1.087	1.155	1.170	1.583	1.527	1.099
NOx_at_EVO_10_5_trial3	1.474	41.443	22.575	7.649	4.138	2.340	8.284	1.135	1.282	1.371	0.132	2.205	1.206	1.113	1.202	1.181	1.562	1.784	1.013
NOx_at_EVO_10_5_5_trial1	1.504	49.990	23.334	2.676	3.728	2.209	9.024	1.098	1.258	1.424	0.117	2.262	1.157	1.096	1.230	1.223	1.551	1.580	1.131
NOx_at_EVO_10_5_5_trial2	1.597	38.645	24.107	7.367	3.698	2.340	8.837	1.095	1.242	1.403	0.128	2.088	1.161	1.026	1.209	1.198	1.600	1.657	1.023
NOx_at_EVO_10_5_5_trial3	1.455	42.305	19.161	2.463	3.605	2.356	8.524	1.103	1.237	1.420	0.136	2.263	1.199	1.073	1.154	1.182	1.577	1.585	1.110
Maximum_Temperature_10_5_trial1	2.095	116.667	88.873	25.645	3.382	9.853	8.611	3.334	3.422	3.016	1.417	5.314	2.355	1.496	1.544	1.413	1.540	2.781	4.367
Maximum_Temperature_10_5_trial2	2.039	107.479	103.000	6.216	3.756	9.894	8.419	3.874	3.492	2.936	1.893	5.508	2.420	1.499	1.550	1.338	1.562	2.860	2.725
Maximum_Temperature_10_5_trial3	2.040	95.267	89.859	36.325	3.678	9.922	8.543	3.295	3.421	3.363	1.410	5.507	2.370	1.506	1.551	1.364	1.544	2.798	4.381
Maximum_Temperature_10_5_5_trial1	1.996	107.326	106.868	6.826	3.454	9.801	8.857	3.349	3.285	3.036	1.850	5.488	2.378	1.531	1.530	1.339	1.570	2.686	4.390
Maximum_Temperature_10_5_5_trial2	2.048	104.201	95.015	48.964	3.484	9.888	8.564	3.326	3.319	2.998	1.423	5.279	2.368	1.447	1.612	1.382	1.475	2.772	4.435
Maximum_Temperature_10_5_5_trial3	2.005	116.055	96.438	22.165	3.317	9.926	8.047	3.340	3.281	2.942	1.836	5.385	2.393	1.470	1.496	1.397	1.478	2.874	4.369

Metamodels R^2

Table 19. R-Squared for all initial metamodels studied

Metamodel \ rpm	1000	1250	1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500
Brake_Torque_Kriging	0.997	0.545	0.598	0.802	0.617	0.020	0.248	0.994	0.997	0.997	1.000	0.957	0.996	0.998	0.995	0.996	0.994	0.995	0.995
Brake_Torque_Linear	0.839	0.301	0.243	0.233	0.554	0.182	0.120	0.943	0.941	0.939	0.935	0.889	0.925	0.956	0.952	0.960	0.941	0.919	0.950
Brake_Torque_Quadratic	0.972	0.550	0.522	0.561	0.634	0.225	0.247	0.997	0.994	0.995	0.996	0.946	0.993	0.995	0.994	0.995	0.980	0.979	0.991
Brake_Torque_Cubic	0.997	0.517	0.540	0.554	0.641	0.217	0.256	0.998	0.998	0.998	1.000	0.948	0.998	0.999	0.998	0.998	0.986	0.984	0.997
Brake_Torque_MLP	0.999	0.697	0.759	0.987	0.699	0.301	0.306	0.998	0.998	0.999	1.000	0.983	0.999	0.999	0.999	0.999	0.998	0.998	0.998
Brake_Torque_IDW	0.871	0.519	0.578	0.818	0.586	0.136	0.282	0.897	0.897	0.902	0.928	0.853	0.908	0.912	0.909	0.932	0.884	0.883	0.914
Brake_Power_Kriging	0.998	0.544	0.612	0.803	0.618	0.039	0.249	0.995	0.996	0.996	1.000	0.961	0.994	0.998	0.995	0.995	0.994	0.994	0.993
Brake_Power_Linear	0.839	0.301	0.243	0.233	0.554	0.182	0.120	0.943	0.941	0.939	0.935	0.889	0.925	0.956	0.952	0.960	0.941	0.919	0.950
Brake_Power_Quadratic	0.972	0.550	0.521	0.561	0.634	0.225	0.246	0.997	0.994	0.995	0.996	0.946	0.993	0.995	0.994	0.995	0.980	0.979	0.991
Brake_Power_Cubic	0.998	0.540	0.544	0.555	0.642	0.215	0.255	0.998	0.998	0.998	1.000	0.948	0.998	0.999	0.998	0.998	0.986	0.984	0.997
Brake_Power_MLP	0.999	0.670	0.758	0.895	0.696	0.236	0.312	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.998	0.997	0.999
Brake_Power_IDW	0.871	0.519	0.578	0.818	0.586	0.136	0.282	0.897	0.897	0.902	0.928	0.853	0.908	0.912	0.909	0.932	0.884	0.883	0.914
Volumentric_efficiency_Kriging	0.999	0.561	0.628	0.788	0.958	0.948	0.548	0.999	1.000	1.000	1.000	0.976	1.000	1.000	1.000	1.000	1.000	0.998	0.999
Volumentric_efficiency_Linear	0.855	0.350	0.275	0.209	0.861	0.941	0.261	0.936	0.923	0.940	0.941	0.877	0.912	0.942	0.936	0.934	0.935	0.932	0.947
Volumentric_efficiency_Quadratic	0.974	0.580	0.547	0.546	0.968	0.978	0.412	0.995	0.993	0.994	0.995	0.954	0.991	0.993	0.994	0.994	0.987	0.985	0.993
Volumentric_efficiency_Cubic	0.998	0.577	0.568	0.540	0.977	0.977	0.423	0.997	0.999	0.999	0.999	0.959	0.998	0.999	0.999	0.998	0.992	0.990	0.998
Volumentric_efficiency_MLP	1.000	0.727	0.747	1.000	0.987	0.985	0.666	0.999	1.000	1.000	1.000	0.984	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumentric_efficiency_IDW	0.876	0.541	0.593	0.813	0.832	0.861	0.478	0.876	0.910	0.891	0.915	0.870	0.912	0.897	0.895	0.898	0.861	0.874	0.887
EGR_Percentage_Kriging	1.000	0.548	0.607	0.817	0.930	0.768	0.319	0.999	1.000	1.000	0.999	0.991	0.999	0.999	1.000	1.000	1.000	0.999	0.999
EGR_Percentage_Linear	0.845	0.295	0.239	0.258	0.931	0.882	0.199	0.926	0.914	0.889	0.883	0.862	0.852	0.855	0.900	0.881	0.830	0.905	0.891
EGR_Percentage_Quadratic	0.983	0.552	0.514	0.572	0.963	0.908	0.348	0.993	0.990	0.988	0.987	0.979	0.982	0.983	0.989	0.989	0.979	0.987	0.985
EGR_Percentage_Cubic	0.999	0.530	0.528	0.565	0.966	0.908	0.344	0.998	0.998	0.997	0.996	0.990	0.996	0.996	0.997	0.997	0.994	0.994	0.995
EGR_Percentage_MLP	1.000	0.824	0.722	0.935	0.972	0.907	0.414	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_IDW	0.869	0.518	0.572	0.823	0.821	0.827	0.408	0.862	0.895	0.887	0.910	0.848	0.861	0.903	0.886	0.879	0.881	0.884	0.906
Engine_AFR_Kriging	0.996	0.532	0.220	0.234	0.316	0.016	0.510	0.706	0.281	0.654	0.680	0.816	0.077	0.795	0.888	0.961	0.975	0.818	0.543
Engine_AFR_Linear	0.951	0.284	0.453	0.143	0.248	0.043	0.233	0.813	0.570	0.669	0.801	0.401	0.203	0.354	0.122	0.098	0.153	0.050	0.311
Engine_AFR_Quadratic	0.996	0.536	0.447	0.287	0.287	0.019	0.354	0.884	0.525	0.726	0.896	0.725	0.139	0.347	0.289	0.634	0.590	-0.117	0.353
Engine_AFR_Cubic	0.997	0.531	0.417	0.322	0.297	0.007	0.354	0.887	0.519	0.715	0.892	0.727	0.065	0.354	0.293	0.622	0.571	-0.177	0.336
Engine_AFR_MLP	0.998	0.785	0.406	0.499	0.444	0.101	0.662	0.901	0.589	0.735	0.894	0.901	0.160	0.919	0.973	0.992	0.992	0.981	0.934
Engine_AFR_IDW	0.881	0.507	0.414	0.279	0.288	0.031	0.558	0.788	0.487	0.674	0.756	0.737	0.181	0.432	0.547	0.646	0.608	0.266	0.292
Cyl1_NOx_at_EVO_Kriging	0.991	0.501	0.288	0.868	0.890	0.888	0.446	0.937	0.593	0.870	0.999	0.895	0.975	0.975	0.969	0.940	0.950	0.942	0.966
Cyl1_NOx_at_EVO_Linear	0.827	0.249	0.074	0.555	0.862	0.928	0.283	0.869	0.538	0.875	0.935	0.905	0.890	0.953	0.939	0.905	0.747	0.733	0.848
Cyl1_NOx_at_EVO_Quadratic	0.972	0.519	0.122	0.756	0.921	0.957	0.317	0.963	0.706	0.949	0.990	0.940	0.966	0.984	0.978	0.974	0.882	0.882	0.964
Cyl1_NOx_at_EVO_Cubic	0.979	0.501	0.138	0.756	0.926	0.956	0.313	0.964	0.738	0.947	0.994	0.942	0.977	0.987	0.985	0.975	0.886	0.885	0.971
Cyl1_NOx_at_EVO_MLP	0.996	0.729	0.478	0.976	0.946	0.961	0.568	0.967	0.759	0.948	1.000	0.956	0.991	0.993	0.989	0.988	0.975	0.983	0.991
Cyl1_NOx_at_EVO_IDW	0.858	0.485	0.326	0.815	0.797	0.868	0.516	0.858	0.669	0.773	0.852	0.771	0.846	0.841	0.857	0.846	0.819	0.724	0.896
Cyl1_Max_Temp_Kriging	0.994	0.537	0.639	0.812	0.765	0.220	0.291	0.148	0.535	0.618	0.903	0.508	0.899	0.928	0.921	0.918	0.950	0.863	0.745
Cyl1_Max_Temp_Linear	0.882	0.317	0.282	0.306	0.918	0.227	0.206	0.499	0.691	0.760	0.827	0.559	0.802	0.808	0.859	0.821	0.739	0.787	0.709
Cyl1_Max_Temp_Quadratic	0.983	0.559	0.553	0.597	0.914	0.232	0.219	0.553	0.775	0.879	0.934	0.640	0.942	0.957	0.968	0.973	0.931	0.914	0.799
Cyl1_Max_Temp_Cubic	0.997	0.550	0.575	0.591	0.921	0.218	0.219	0.559	0.816	0.873	0.932	0.641	0.951	0.970	0.976	0.983	0.954	0.923	0.822
Cyl1_Max_Temp_MLP	0.998	0.639	0.767	0.872	0.912	0.256	0.470	0.553	0.799	0.875	0.963	0.662	0.955	0.980	0.985	0.988	0.984	0.950	0.832
Cyl1_Max_Temp_IDW	0.882	0.525	0.598	0.829	0.832	0.243	0.439	0.468	0.666	0.672	0.850	0.546	0.806	0.861	0.848	0.849	0.810	0.806	0.751

Table 20. MLP metamodels R-Squared

Metamodel \ rpm	1000	1250	1500	1750	2000	2250	2500	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000	5250	5500
Brake_Torque_10_5_trial1	0.999	0.683	0.764	0.965	0.700	0.255	0.338	0.998	0.998	0.999	1.000	0.981	0.999	0.999	0.999	0.999	0.998	0.998	0.999
Brake_Torque_10_5_trial2	0.999	0.702	0.817	0.916	0.706	0.286	0.326	0.998	0.998	0.999	1.000	0.985	0.999	0.999	0.999	0.999	0.998	0.998	0.998
Brake_Torque_10_5_trial3	0.999	0.618	0.751	0.805	0.691	0.275	0.305	0.998	0.998	0.999	1.000	0.982	0.999	0.999	0.999	0.999	0.999	0.998	0.999
Brake_Torque_10_5_5_trial1	0.999	0.822	0.754	0.901	0.699	0.286	0.331	0.998	0.998	0.999	1.000	0.983	0.999	0.999	0.999	0.999	0.999	0.998	0.999
Brake_Torque_10_5_5_trial2	0.999	0.699	0.768	0.944	0.685	0.343	0.287	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.999	0.998	0.999
Brake_Torque_10_5_5_trial3	0.999	0.768	0.767	0.967	0.675	0.279	0.344	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.999	0.998	0.998
Brake_Power_10_5_trial1	0.999	0.760	0.837	0.999	0.682	0.235	0.110	0.998	0.998	0.999	1.000	0.983	0.999	0.999	0.999	0.999	0.998	0.998	0.999
Brake_Power_10_5_trial2	0.999	0.711	0.830	0.956	0.724	0.361	0.298	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.998	0.998	0.998
Brake_Power_10_5_trial3	0.999	0.711	0.784	0.999	0.725	0.167	0.303	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.998	0.998	0.999
Brake_Power_10_5_5_trial1	1.000	0.869	0.757	0.935	0.719	0.262	0.300	0.998	0.998	0.999	1.000	0.983	0.999	0.999	0.999	0.999	0.998	0.998	0.998
Brake_Power_10_5_5_trial2	1.000	0.707	0.801	0.961	0.675	0.281	0.325	0.998	0.998	0.999	1.000	0.984	0.999	0.999	0.999	0.999	0.998	0.998	0.998
Brake_Power_10_5_5_trial3	0.999	0.736	0.774	0.996	0.680	0.279	0.263	0.998	0.998	0.999	1.000	0.983	0.999	0.999	0.999	0.999	0.999	0.998	0.998
Volumetric_Efficiency_10_5_trial1	1.000	0.776	0.765	0.924	0.987	0.986	0.642	0.999	1.000	1.000	1.000	0.988	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumetric_Efficiency_10_5_trial2	1.000	0.730	0.768	0.999	0.984	0.987	0.611	0.999	1.000	1.000	1.000	0.986	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumetric_Efficiency_10_5_trial3	1.000	0.698	0.747	0.942	0.986	0.985	0.572	0.999	1.000	1.000	1.000	0.984	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumetric_Efficiency_10_5_5_trial1	1.000	0.866	0.798	1.000	0.988	0.983	0.571	0.999	1.000	1.000	1.000	0.986	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumetric_Efficiency_10_5_5_trial2	1.000	0.728	0.810	0.955	0.988	0.986	0.599	0.999	1.000	1.000	1.000	0.985	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Volumetric_Efficiency_10_5_5_trial3	1.000	0.710	0.723	0.814	0.987	0.986	0.619	0.999	1.000	1.000	1.000	0.986	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_trial1	1.000	0.662	0.840	0.996	0.975	0.914	0.443	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_trial2	1.000	0.823	0.764	1.000	0.975	0.908	0.403	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_trial3	1.000	0.674	0.726	0.999	0.972	0.911	0.330	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_5_trial1	1.000	0.756	0.788	0.998	0.974	0.917	0.425	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_5_trial2	1.000	0.745	0.837	0.991	0.973	0.912	0.438	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000
EGR_Percentage_10_5_5_trial3	1.000	0.753	0.733	0.999	0.974	0.917	0.440	1.000	1.000	1.000	1.000	0.994	1.000	1.000	1.000	1.000	1.000	1.000	1.000
AFR_10_5_trial1	0.998	0.851	0.466	0.489	0.516	0.084	0.667	0.901	0.472	0.735	0.891	0.897	0.135	0.913	0.971	0.991	0.995	0.984	0.899
AFR_10_5_trial2	0.998	0.709	0.472	0.499	0.482	0.081	0.610	0.887	0.520	0.734	0.894	0.871	0.203	0.914	0.969	0.992	0.997	0.957	0.921
AFR_10_5_trial3	0.999	0.762	0.461	0.502	0.480	0.141	0.620	0.904	0.549	0.731	0.884	0.898	0.195	0.903	0.973	0.991	0.993	0.963	0.905
AFR_10_5_5_trial1	0.999	0.688	0.497	0.472	0.513	0.111	0.568	0.907	0.594	0.728	0.899	0.894	0.191	0.901	0.968	0.993	0.998	0.967	0.903
AFR_10_5_5_trial2	0.999	0.656	0.541	0.471	0.453	0.104	0.608	0.905	0.588	0.755	0.893	0.896	0.196	0.892	0.971	0.992	0.990	0.964	0.931
AFR_10_5_5_trial3	0.999	0.675	0.507	0.512	0.500	0.157	0.564	0.908	0.591	0.739	0.897	0.880	0.195	0.920	0.974	0.992	0.994	0.971	0.930
NOx_at_EVO_10_5_trial1	0.997	0.795	0.444	0.989	0.942	0.960	0.617	0.971	0.767	0.949	1.000	0.963	0.991	0.993	0.989	0.989	0.977	0.987	0.993
NOx_at_EVO_10_5_trial2	0.997	0.786	0.500	0.980	0.946	0.961	0.473	0.968	0.768	0.951	1.000	0.959	0.991	0.993	0.989	0.988	0.973	0.983	0.992
NOx_at_EVO_10_5_trial3	0.997	0.729	0.515	0.950	0.935	0.961	0.612	0.970	0.757	0.950	1.000	0.959	0.991	0.992	0.988	0.988	0.974	0.977	0.993
NOx_at_EVO_10_5_5_trial1	0.996	0.605	0.482	0.994	0.947	0.965	0.539	0.971	0.766	0.947	1.000	0.956	0.992	0.992	0.987	0.987	0.974	0.982	0.991
NOx_at_EVO_10_5_5_trial2	0.996	0.764	0.448	0.953	0.948	0.961	0.558	0.972	0.772	0.948	1.000	0.963	0.992	0.993	0.987	0.987	0.972	0.980	0.993
NOx_at_EVO_10_5_5_trial3	0.997	0.717	0.651	0.995	0.951	0.960	0.589	0.971	0.774	0.947	1.000	0.956	0.991	0.993	0.989	0.988	0.973	0.982	0.991
Maximum_Temperature_10_5_trial1	0.998	0.655	0.816	0.968	0.926	0.260	0.406	0.545	0.810	0.869	0.980	0.689	0.959	0.980	0.985	0.987	0.984	0.951	0.827
Maximum_Temperature_10_5_trial2	0.998	0.707	0.753	0.998	0.909	0.254	0.432	0.386	0.802	0.876	0.963	0.665	0.956	0.980	0.985	0.988	0.984	0.948	0.933
Maximum_Temperature_10_5_trial3	0.998	0.770	0.812	0.936	0.913	0.249	0.415	0.556	0.810	0.837	0.980	0.666	0.958	0.980	0.985	0.988	0.984	0.950	0.826
Maximum_Temperature_10_5_5_trial1	0.998	0.708	0.734	0.998	0.923	0.267	0.371	0.541	0.825	0.867	0.965	0.668	0.958	0.979	0.985	0.988	0.984	0.954	0.826
Maximum_Temperature_10_5_5_trial2	0.998	0.725	0.789	0.884	0.922	0.254	0.412	0.547	0.821	0.871	0.979	0.693	0.958	0.981	0.984	0.988	0.986	0.951	0.822
Maximum_Temperature_10_5_5_trial3	0.998	0.658	0.783	0.976	0.929	0.249	0.481	0.543	0.825	0.876	0.966	0.680	0.957	0.981	0.986	0.987	0.986	0.948	0.827